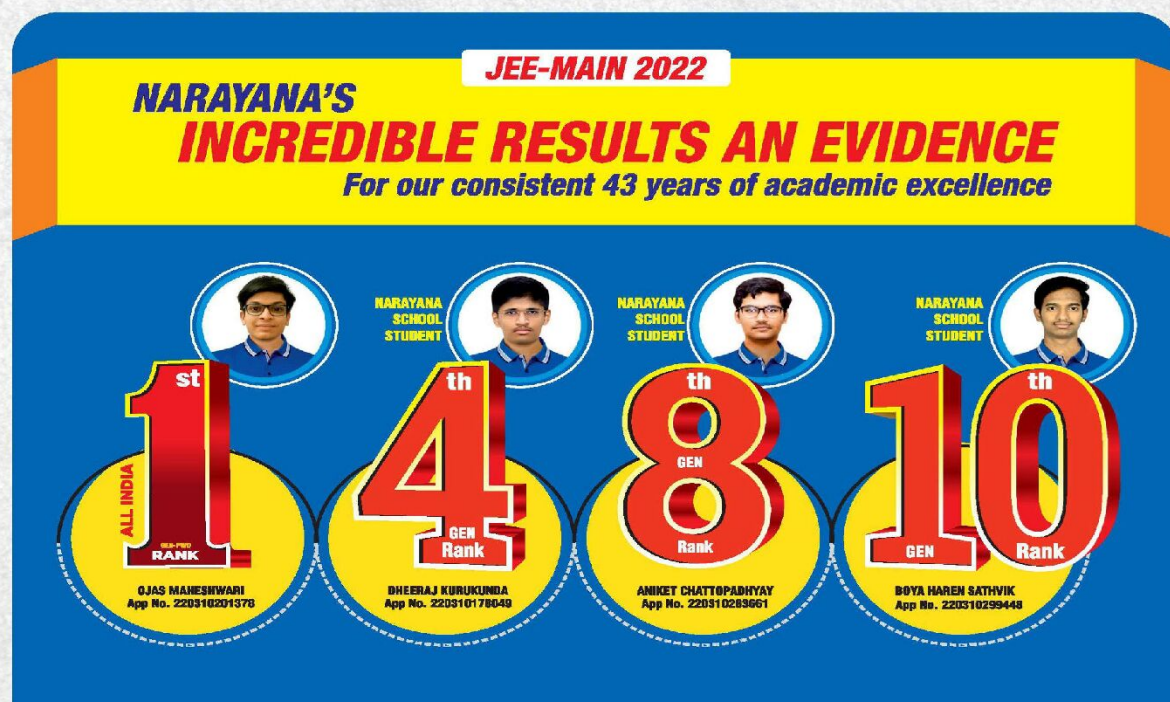




NARAYANA
IIT-JEE ACADEMY - INDIA

NARAYANA GRABS THE LION'S SHARE IN JEE-ADV.2022
EXCELLENCE IS OUR TRADITION 3402 RANKS OVERALL!



CHEMISTRY

ORGANIC CHEMISTRY PART-I(JR)
JEE MAIN IMPORTANT QUESTIONS



NARAYANA
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ORGANIC CHEMISTRY PART-I(JR)

Single Answer Type Questions

01. Choose the incorrect statement.

- A) C_6H_6 is more reactive than C_6D_6 towards nitration.
 B) C_6D_6 is more reactive than C_6T_6 towards sulphonation.
 C) C_6H_6 is more reactive than C_6D_6 towards sulphonation.
 D) C_6D_6 and C_6H_6 have equal reactivity towards nitration

Key. A

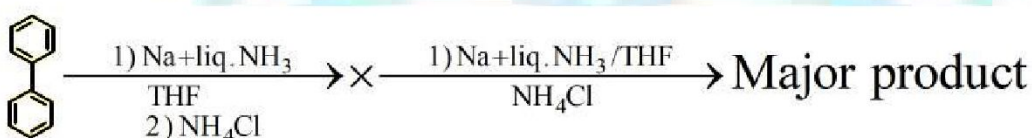
Sol. $K_H/K_D = 1$ for nitration of benzene

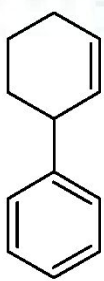
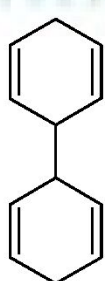
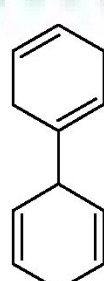
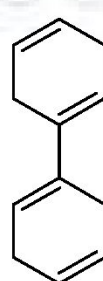
02. A solution of cyclohexene in benzene is stirred at $0^\circ C$, while concentrated sulphuric acid is added, after washing away the acid and removing excess benzene, what product is obtained?

- A) 1-phenylcyclohexene B) cyclohexyl benzene
 C) Trans-1, 2-diphenyl cyclohexane D) 1,1-diphenyl cyclohexane

Key. B

Sol. Cyclohexene produces cyclohexyl cation with acid which is then attacked by benzene to form cyclohexyl benzene

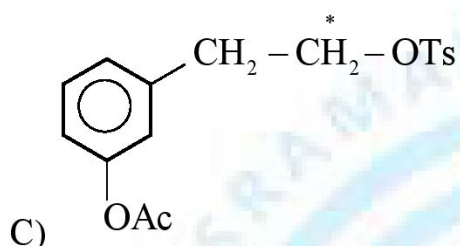
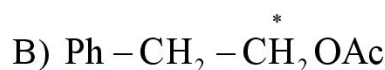
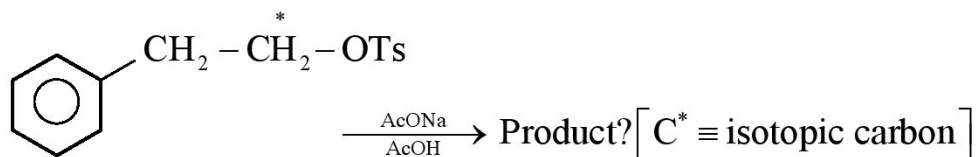
03.  Major product

- A)  B)  C)  D) 

Key. C

Sol. Alkyl substituted double bond is retained during Birch reduction

04.



D) Both A and B

Key. D

Sol. Neighbouring group participation

05. **Assertion:** When aniline is subjected to nitration by conc. HNO_3 and H_2SO_4 meta nitro aniline is formed in considerable amount.

Reason: NH_2 is o/p directing but ring deactivating group.

A) Assertion is True. Reason is true: Reason is the correct explanation for Assertion.

B) Assertion is True Reason is true: Reason is not the correct explanation for Assertion.

C) Assertion is True, Reason is False.

D) Assertion is False, Reason is true.

Key. C

Sol. Refer NCERT volume II 12th Page 395.

06. **Assertion:** Benzene does not undergo addition reaction easily.

Reason: Benzene loses its aromaticity after addition reaction & becomes less stable.

A) Assertion is True, Reason is true. Reason is the correct explanation for Assertion.

B) Assertion is True, Reason is true. Reason is not the correct explanation for Assertion.

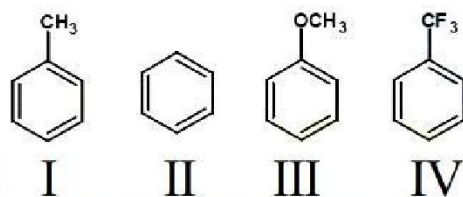
C) Assertion is True, Reason is False.

D) Assertion is False, Reason is true.

Key. A

Sol. Aromatic character of benene prevents it from undergoing addition reaction

07. Among the compounds the order of decreasing reactivity towards electrophilic aromatic substitution is:



A) II > I > III > IV B) III > I > II > IV

C) III > I > IV > II D) II > I > IV > III

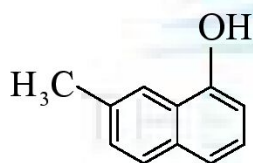
Key. B

Sol. Reactivity towards electrophilic aromatic substitution $\propto$ electron density of benzene ring

08. In a chemical reaction, the compound (X).



can rather conveniently be converted to another compound(Y)



7-methyl 1-naphthol

Which of the following can be most suitable first step for this purpose of conversion?

A) Trans isomer of (X) can be treated with anhydrous AlCl_3

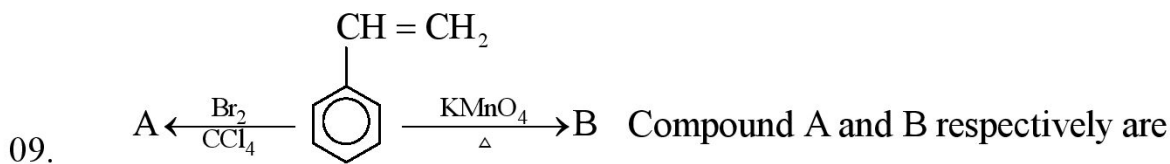
B) Cis isomer of (X) can be treated with $\text{HF} + \text{BF}_3$

C) Trans isomer of (X) can be treated with cone. H_3PO_4

D) Cis isomer of (X) can be treated with AlCl_3

Key. B

Sol. In the process Step -1 $\rightarrow$ Formation of acyl cation, Step -2 $\rightarrow$ EAS at meta position to methyl group, Step -3 $\rightarrow$ Enolisation to get aromatic stabilised phenol

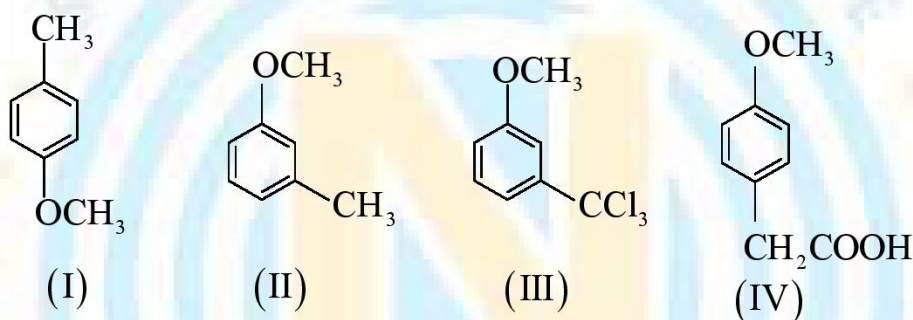


- A) o-bromo styrene, benzoic acid B) p-bromo styrene, benzaldehyde
C) m-bromo styrene, benzaldehyde D) styrene dibromide, benzoic acid

Key. D

Sol.

10. The order of reactivity of the following compounds in electrophilic monochlorination at the most favourable position is



- A) I < II < IV < III B) III < IV < I < II
C) IV < III < II < I D) III < II < IV < I

Key. B

Sol. Reactivity towards electrophilic aromatic substitution $\propto$ electron density of benzene ring

11. Incorrect statement regarding Friedel-Crafts alkylation reaction is

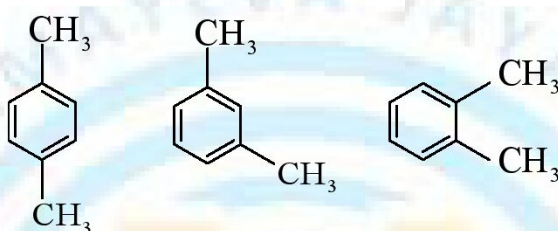
- A) When $\text{CH}_3\text{Br}/\text{AlBr}_3$ is used for alkylation of benzene, the electrophile involved is free methyl carbocation.
B) As alkyl groups are activating in nature, Friedel-Crafts reaction leads to poly alkylation.
C) When $\text{Me}_3\text{CCH}_2\text{Cl}/\text{FeCl}_3$ is used for alkylation of benzene, unrearranged product is formed almost exclusively.

D) Low temperatures facilitate unrearranged product.

Key. A

Sol. With primary alkyl halides free carbocations are not formed as they are unstable. Weak Lewis acids and low temperatures facilitate unrearranged products.

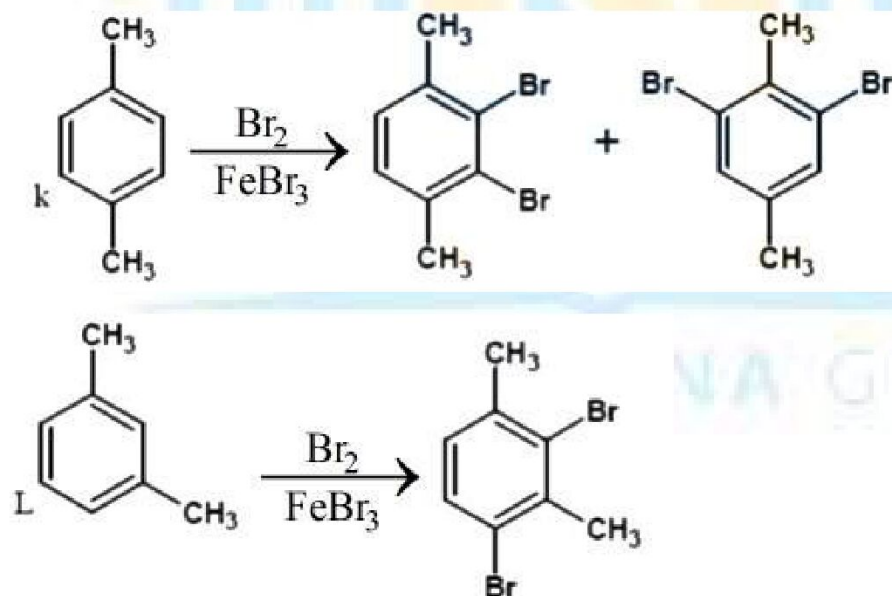
12. Which of the following xylene(s) can give 1,3-dibromo product as one of the possible product (as per IUPAC nomenclature) on reaction with $\text{Br}_2/\text{FeBr}_3$. (Hint: Consider methyl group, CH_3 as ortho / para directing group use dimethyl benzene as suffix, don't use xylene as suffix)

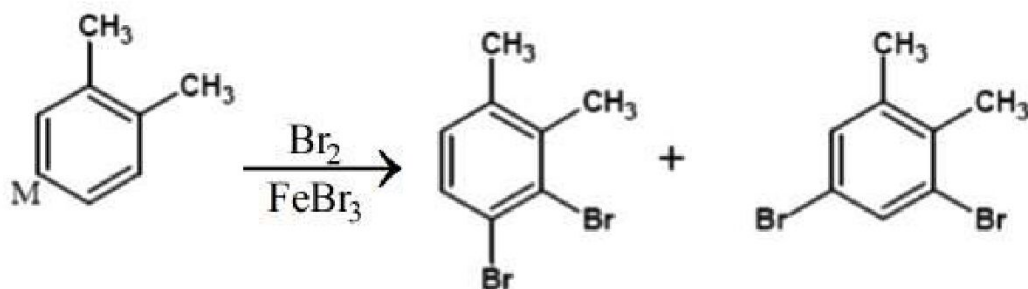


- A) K & L both B) L only C) K & M both D) L & M both

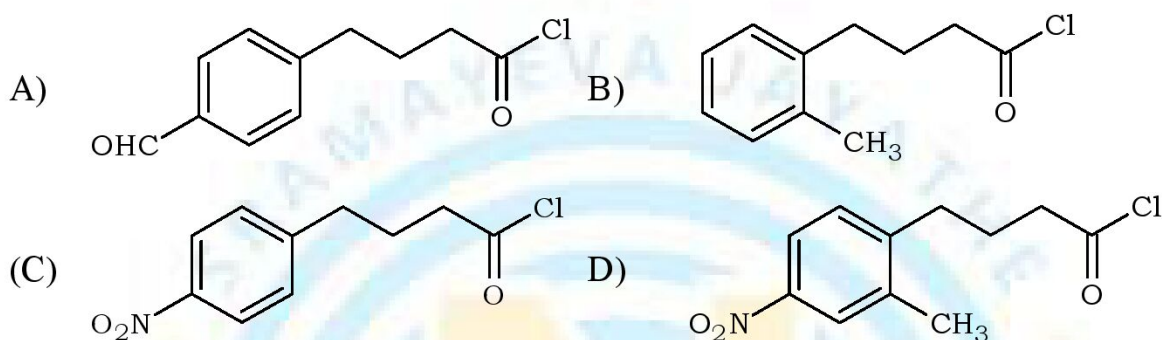
Key. A

Sol.





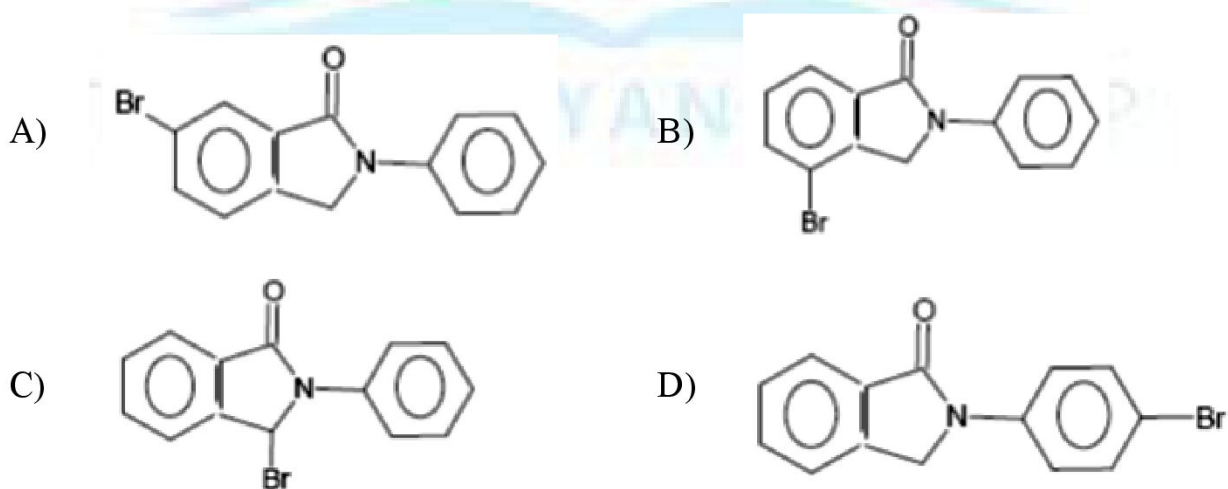
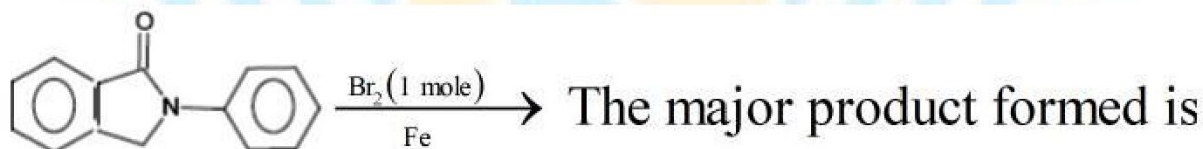
13. Which of the following compound cyclizes most readily when treated with anhydrous AlCl_3 ?



Key. B

Sol. Presence of deactivating groups do not facilitate Friedel Craft's reactions.

14. In the reaction:



Key. D

Sol. Nitrogen lone pair in conjugation with the benzene ring makes, para position more reactive for bromination.

15. Which of the following option(s) is **correct**?

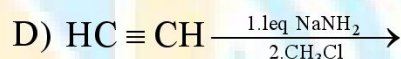
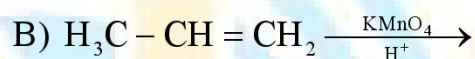
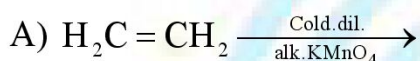


D) all of them

Key. A

Sol. Refer 11th NCERT part II page 382

16. In which of the following reactions, obtained organic product cannot show conformational isomerism?



Key. D

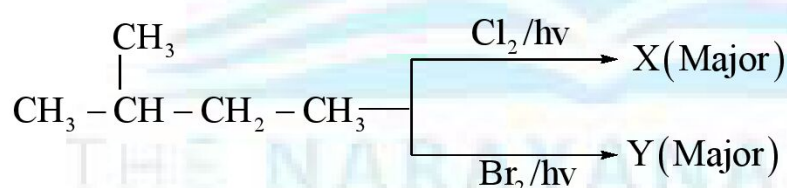
Sol. A) Product is $\text{CH}_2\text{OH}-\text{CH}_2\text{OH}$

B) Product is $\text{CH}_3\text{CHOH}-\text{CH}_2\text{OH}$

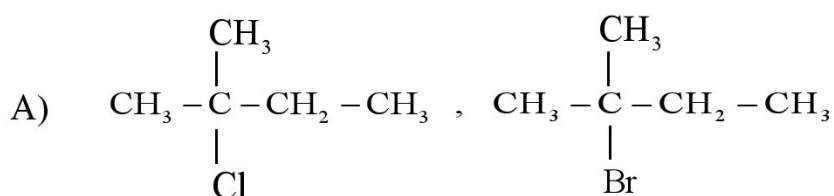
C) Product is $\text{CH}_2\text{Br}-\text{CH}_2\text{Br}$

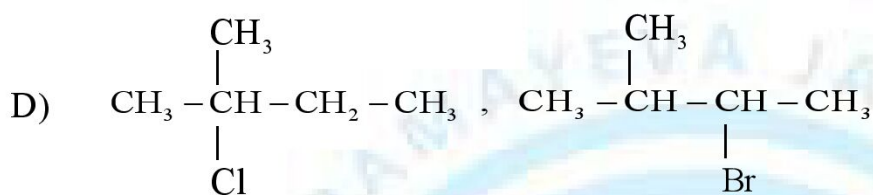
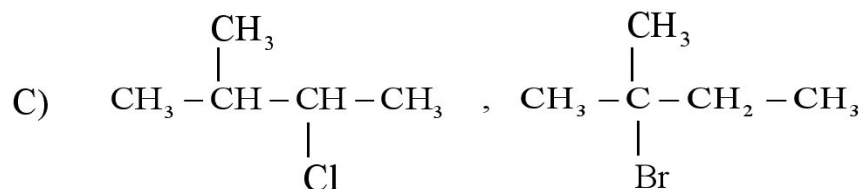
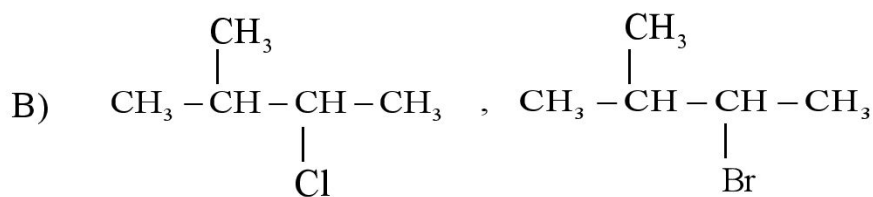
D) Product is $\text{HC}\equiv\text{CCH}_3$

17.



X and Y are respectively.

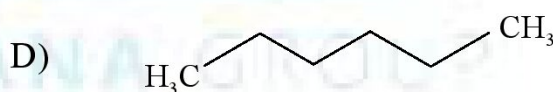
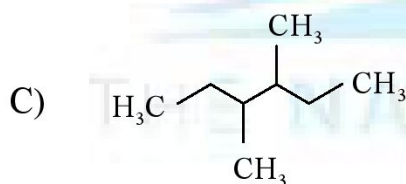
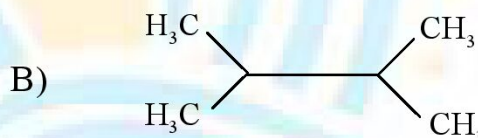
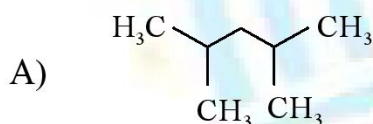
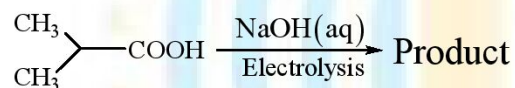




Key. C

Sol. Bromination is much more regioselective than chlorination

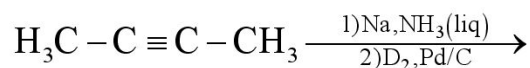
18. The end product of the reaction sequence is

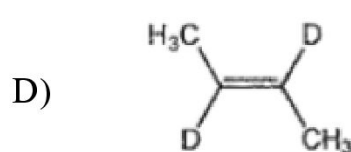
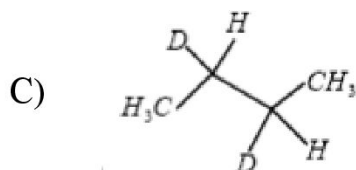
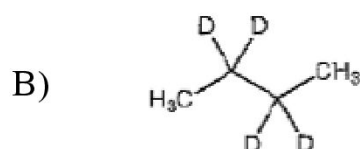
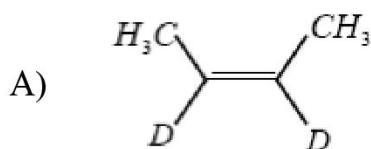


Key. B

Sol. It is Kolbe's electrolysis followed by free radical substitution which is followed by Wurtz reaction.

19. What is the major organic product of the following reaction?

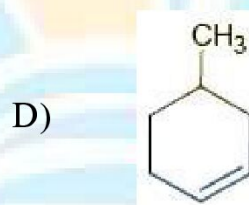
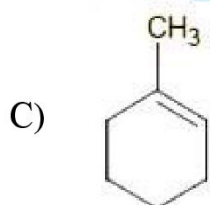
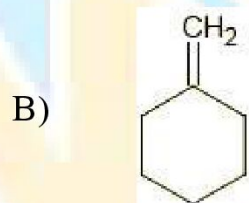
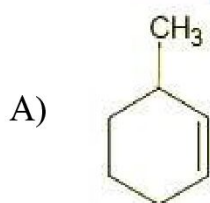
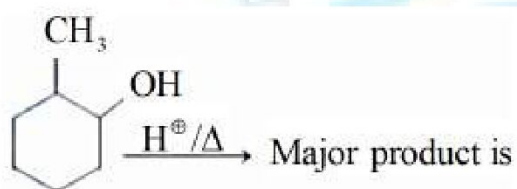




Key. C

Sol. Follow the rules for stability of resonating structures (Structure with more number of π -bond is more stable)

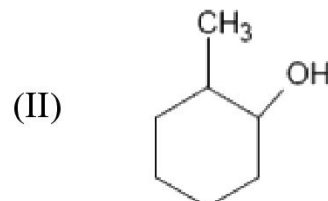
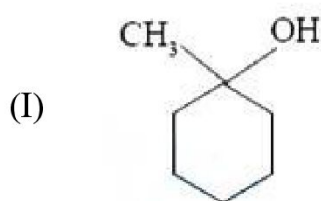
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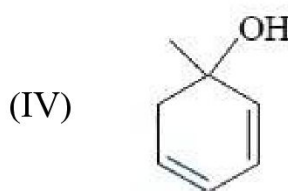
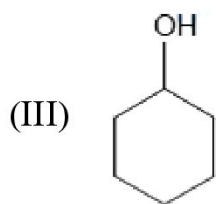


Key. C

Sol. Dehydration of alcohol follow E1 mechanism and zaitsev rule.

21. Arrange following alcohols in decreasing order of their reactivity for dehydration reaction in acidic medium





A) IV > II > III > I

B) IV > I > II > III

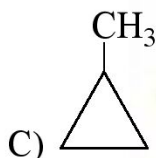
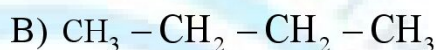
C) I > III > II > IV

D) II > III > IV > I

Key. B

Sol. Greater than stability of carbocation faster is the rate of dehydration.

22. Which of the following will not give three or more monochloro structural products after monochlorination?



Key. B

Sol. Option (B) has only 2 different set of equivalent hydrogen. That is why only two monochloro products are possible.

23. HBr reacts fastest with

A) 1-butene

B) 1-pentane

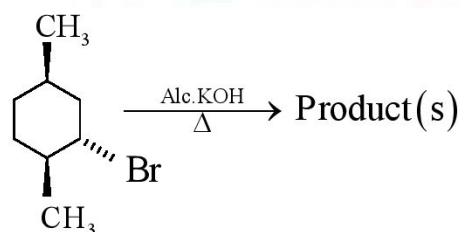
C) 2-pentane

D) 2-methyl propene

Key. D

Sol. Conceptual

- 24.



Find the number of product (s) (including stereoisomers)

A) 2

B) 3

C) 4

D) 5

Key. B

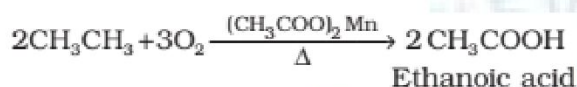
Page | 12

- B) n-butane boils at melting temperature of ice
 C) Neo pentane has '0' secondary carbons and '4' primary carbons
 D) Ethane on oxidation with Manganese acetate gives acetic acid

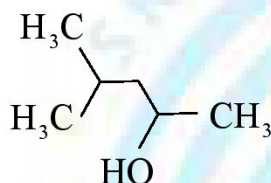
Key. A

Sol. $\text{RCl} - (\text{Zn} + \text{HCl}) \rightarrow \text{RH}$;

B.P. of butane 272.4K



30. Acid catalysed dehydration of this



molecule can't give

- A) 2-methyl pent-2-ene B) 4-methyl pent-1-ene
 C) 4-methyl pent-2-ene D) all of these are formed

Key. A

Sol. Acid catalysed dehydration don't give 3° carbocation.

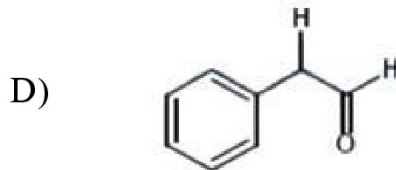
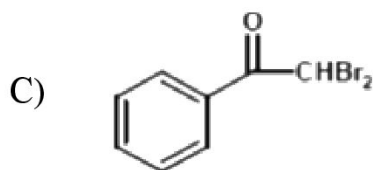
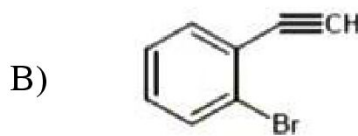
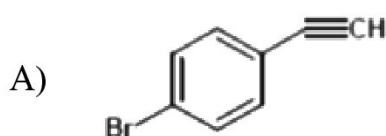
31. Which of the following reaction can be used to differentiate $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ and $\text{CH}=\text{C}-\text{CH}_2-\text{CH}_3$?

- A) $\text{Cu}_2\text{Cl}_2 + \text{NH}_4\text{OH}$ B) dil alkaline KMnO_4 solution
 C) Br_2 water D) HBr

Key. A

Sol. 1-yne gives red precipitate with $\text{Cu}_2\text{Cl}_2 + \text{NH}_4\text{OH}$

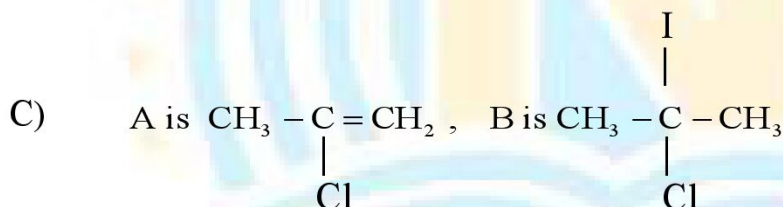
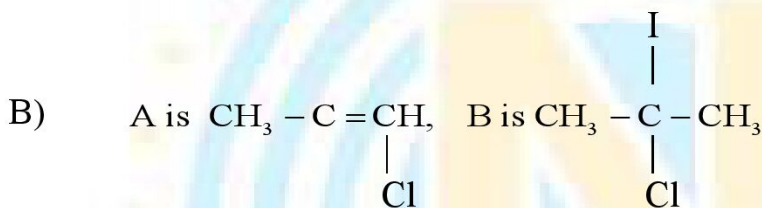
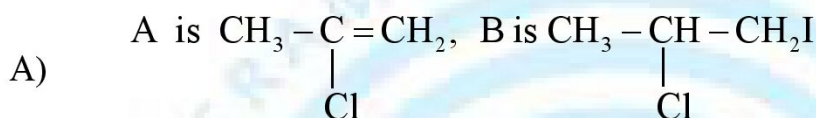
32. $\text{Ph}-\text{C}=\text{CH} \xrightarrow[2\text{eq}]{\text{Br}_2/\text{H}_2\text{O}}$ Product is?



Key. C

Sol. H_2O attack at more substituted carbon.

33. $\text{CH}_3 - \text{C} \equiv \text{CH} \xrightarrow[\text{(1eq)}]{\text{HCl}} \underset{\text{(major)}}{\text{A}} \xrightarrow[\text{(major)}]{\text{HI}} \text{B}$; then find A and B in the reaction sequence?



Key. C

Sol. Electrophilic addition taken place following markownikoff rule.

34. An unknown compound (A) has a molecular formula C_4H_6 . When (A) is treated with excess of Br_2 a new substance (B) with formula. $\text{C}_4\text{H}_6\text{Br}_4$ is formed (A) forms a white ppt. With ammonical silver nitrate solution.(A) may be:

A) But-1-yne B) But-2-yne C) But-1-ene D) But-2-ene

Key. A

Sol. Terminal alkyne reacts with Ammonical silver nitrate sol

35. The melting points of n-pentane (I), isopentane (II) and neopentane (III) increase in the order

A) $II < I < III$ B) $III < II < I$ C) $III < I < II$ D) $I < II < III$

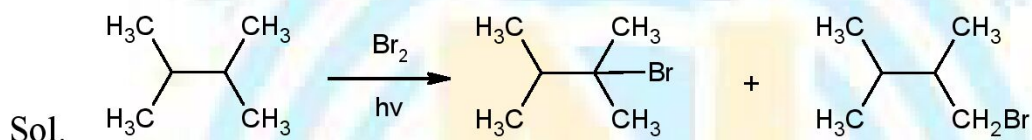
Key. A

Sol. isopentane < n-pentane < neopentane

36. An alkane with molecular mass 86 on bromination gives only two monobromo derivatives (excluding stereoisomers). The alkane is

A) $(CH_3)_2CHCH_2CH_2CH_3$ B) $(CH_3)_2C-CH_2CH_3$
 C) $(CH_3)_2CH-CH(CH_3)_2$ D) $(CH_3)_4C$

Key. C



37. What are X and Y respectively in the following reaction ?

Z-isomer $\xleftarrow{Y}$ 2-butyne $\xrightarrow{X}$ E-isomer

A) $Na/NH_{3(lq)}$ and $Pd/BaSO_4$ poisoned with S + H_2
 B) $Pd/BaSO_4 + H_2$ and $Pd/BaSO_4 + H_2$
 C) Pd/C poisoned with S + H_2 and $Na/NH_{3(lq)}$
 D) Lindlars catalyst and $Na/NH_{3(lq)}$

Key. A

Sol. Pd/C poisoned with S + H_2 and $Na/NH_{3(lq)}$

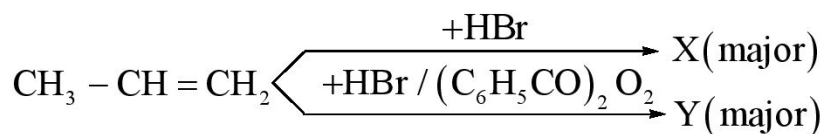
38. When propyne is treated with aqueous H_2SO_4 in the presence of $HgSO_4$, the major product is

A) Propanol B) Propyl hydrogen sulphate
 C) Acetone D) Original mixture

Key. C

Sol. Hydration reaction

39.



Incorrect statement among the following is.

A) X is isopropyl bromide. Y is n-propyl bromide

B) X and Y are positional isomers

C) Formation of X is an electrophilic addition reaction

D) X is n-propyl bromide. Y is isopropyl bromide

Key. D

Sol. X is isopropyl bromide, Y is n-propyl bromide.

40. The treatment of CH_3MgCl with $\text{CH}_3\text{C}\equiv\text{CH}$ produces

A) CH_4

B) $\text{CH}_3 - \text{CH} = \text{CH}_2$

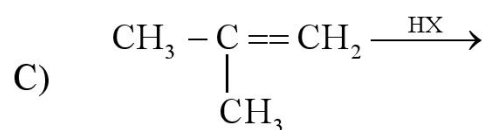
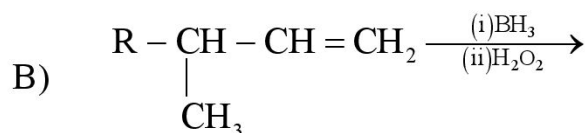
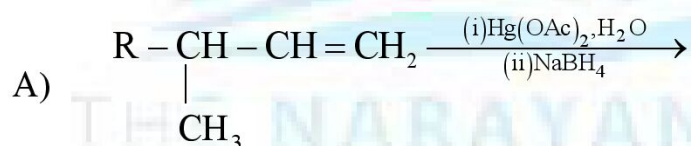
C) $\text{CH}_3\text{C} = \text{C} - \text{CH}_3$

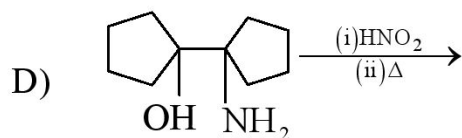
D) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$

Key. A

Sol. RMgX on reaction with Acidic H gives R-H

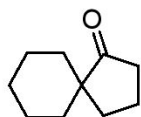
41. Which of the following reaction takes place by formation and rearrangement of carbocation?





Key. D

Sol. Its pinacol-pmacolone type reaction and product is formed after formation of carbocation and its rearrangement by ring expansion.

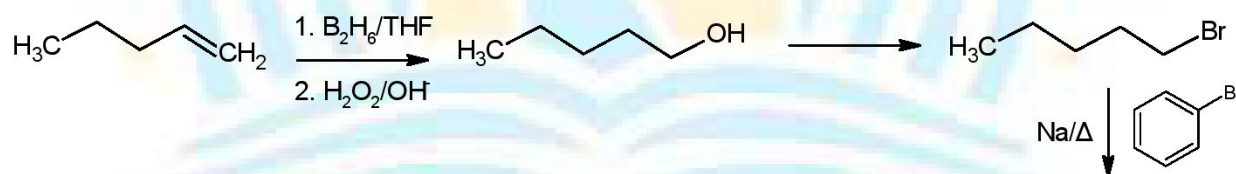


42. Compound "X" reacts with diborane followed by alkaline hydrogen peroxide to form compound Y. "Y" on reaction with a mixture of sodium bromide in sulphuric acid followed by bromobenzene and sodium in ether gives n-pentylbenzene. Compound "X"



Key. D

Sol.



43. $\text{CH}_4 + (\text{O}) \xrightarrow{\text{Cu}/523/100 \text{ atm}} \text{X};$
 $\text{CH}_4 + (\text{O}) \xrightarrow[\Delta]{\text{MO}_2\text{O}_3} \text{Y}$ then X and Y are

- A) $\text{CH}_3\text{OH}, \text{MeOH}$ B) $\text{MeOH}, \text{CH}_3\text{CHO}$
 C) $\text{CH}_3\text{CHO}, \text{HCHO}$ D) $\text{CH}_3\text{OH}, \text{HCHO}$

Key. D

Sol. NCERT Page 374

44. In which of the following reaction new C-C bond is formed?

- A) Decarboxylation of carboxylic acids with soda lime

- B) Reduction of alkyl halides with Zn-Cu in alcohol
 C) Kolbe's electrolysis
 D) All the above

Key. C

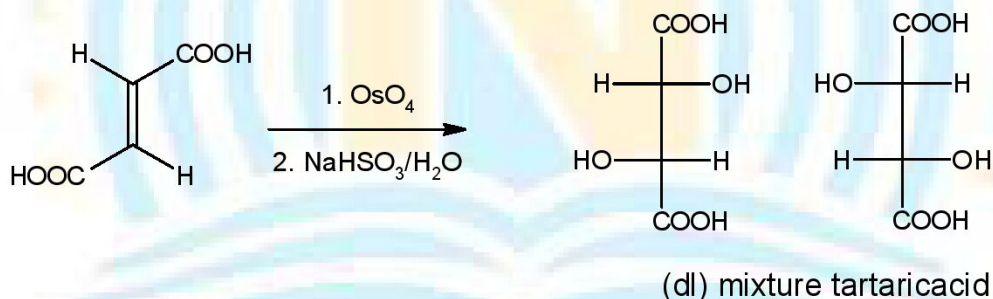
Sol. NCERT 371

45. **Statement-1:** Treatment of fumaric acid with OsO_4 followed by hydrolysis with an aqueous solution of NaHSO_3 gives a mixture of (+) and (-) lactic acids.

Statement-2: OsO_4 brings about syn- addition of hydroxyl groups to the double bond.

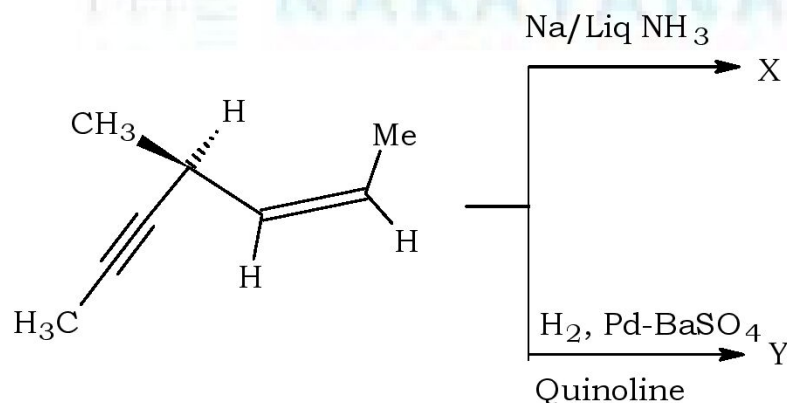
- A) Statement 1 is true. Statement 2 is true: statement 2 is correct explanation for Statement 1.
 B) Statement 1 is true. Statement 2 is true: statement 2 is NOT correct explanation for Statement 1.
 C) Statement 1 is true. Statement 2 is false.
 D) Statement 1 is false. Statement 2 is true.

Key. D



Sol.

46.



The correct statements regarding X and Y is/are

- A) Both X and Y are optically inactive
 B) Both X and Y are optically active
 C) X is optically inactive and Y is active
 D) X is optically active and Y is inactive

Key. D

Sol. Na/liq NH_3 reduces $\text{C}\equiv\text{C}$ to a trans alkene while H_2 , Pd-BaSO₄ reduces $\text{C}\equiv\text{C}$ to a cis alkene

47. The molecules that will have dipole moment is/are

- A) 2,2-dimethyl propane B) trans-2-pentene
 C) Trans-2-butene D) 2,2,3,3-Tetra methyl butane

Key. B

Sol.

48. Three isomeric alkenes A, B and C, (C_5H_{10}) are hydrogenated to yield 2-methyl butane. A and B give the same 3° ROH on oxymercuration-demercuration. B and C give different 1° ROH on hydroboration-oxidation. Some of the possible isomeric structures for C_5H_{10} are as follows:



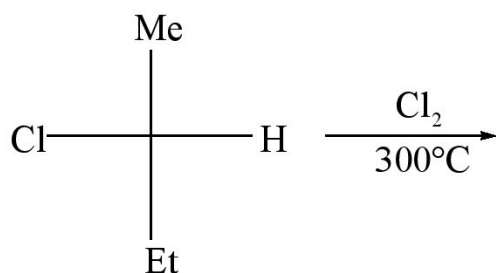
Mark out the correct combination of isomeric A, B and C with structures I to IV.

- A) I-A, II-B, III-C B) II-A, III-B, IV-C
 C) III-A, IV-B, II-C D) III-A, II-B, IV-C

Key. C

Sol. NCERT 383 and second year (vol-ii-322)

49. Pick the correct statement for monochlorination of R-2-chlorobutane



- A) There are four possible products: three are optically active, one is optically inactive
 B) There are five possible products: three are optically inactive & two are optically active.
 C) There are five possible products: two are optically inactive & three are optically active.
 D) There are four possible products: two are optically active & two are optically inactive.

Key. C

Sol. Chlorination of alkane is not enough regioselective that's why produces mixture of dichloroalkanes.

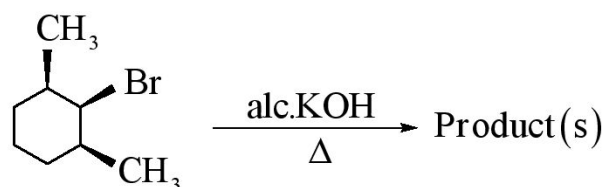
50. $\text{Ph}-\text{CH}_2-\text{CH}_2\text{Br}$ on treatment with $\text{C}_2\text{H}_5\text{OD} / \text{C}_2\text{H}_5\text{O}^-$ gives $\text{Ph}-\text{CD}=\text{CH}_2$ as the product. Which of the following is the most appropriate mechanism for this reaction.

- A) E_1 B) E_2 C) $\text{E}_{1\text{cb}}$ D) E_i

Key. C

Sol. βH exchange with D indicates $\text{E}_{1\text{cb}}$

51.

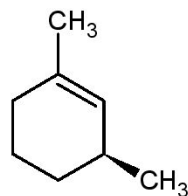
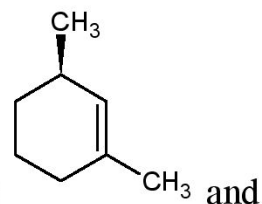


Number of E_2 alkene products (including stereoisomers) is/are

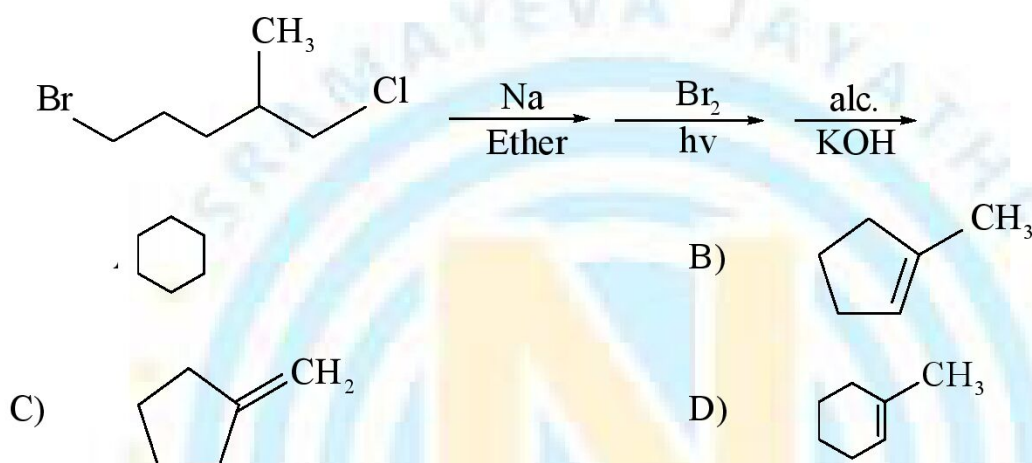
- A) 1 B) 2 C) 3 D) 4

Key. B

Sol. Beta elimination of given compound with alc. KOH produces



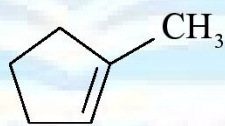
52. The major product of the following reaction is



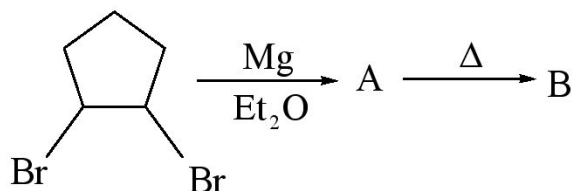
Key. B

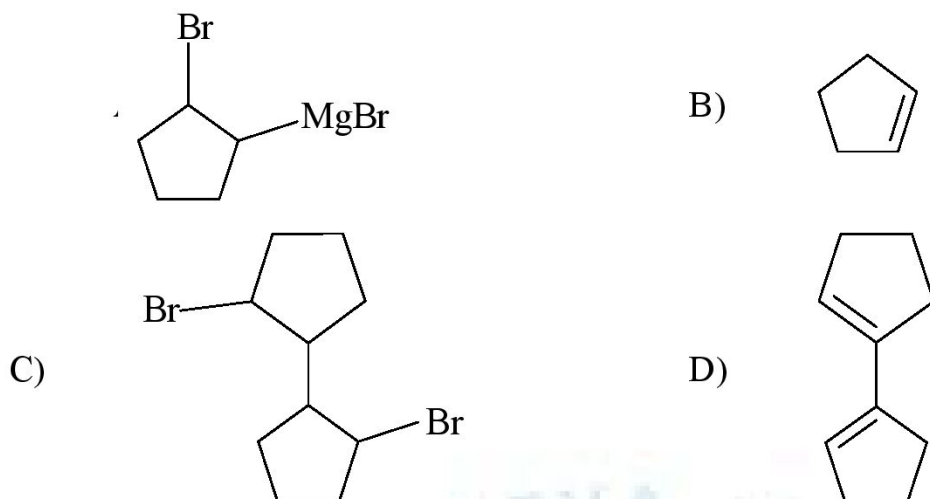
Sol. Step -1 is Wurtz reaction, step -2 is bromination at 3° carbon, step -3 is Beta

elimination with alc. KOH to give



53. In the above reaction, compound B is

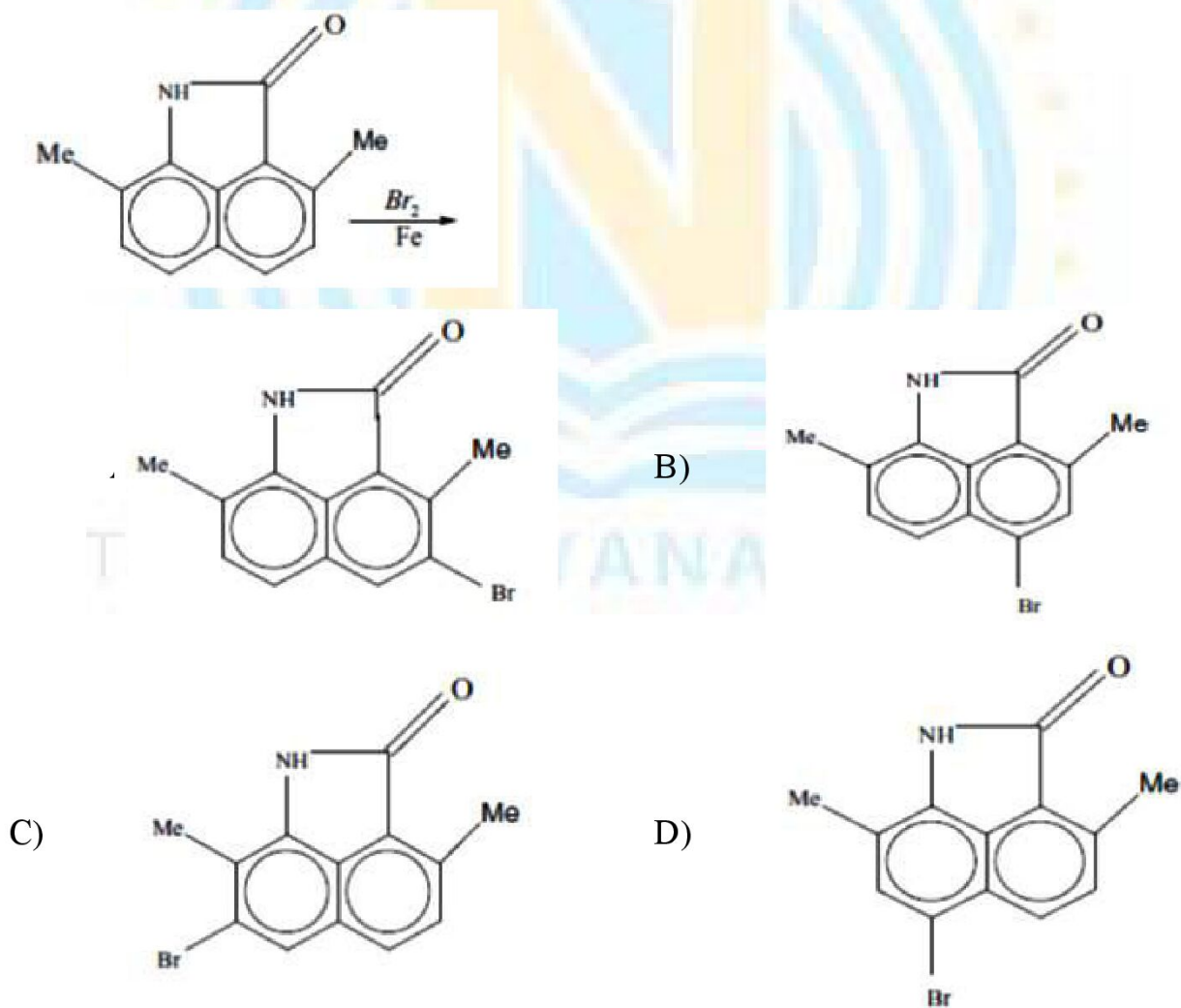




Key. B

Sol. Vicinal dihaloalkanes on reaction with Mg/Ether gives alkene

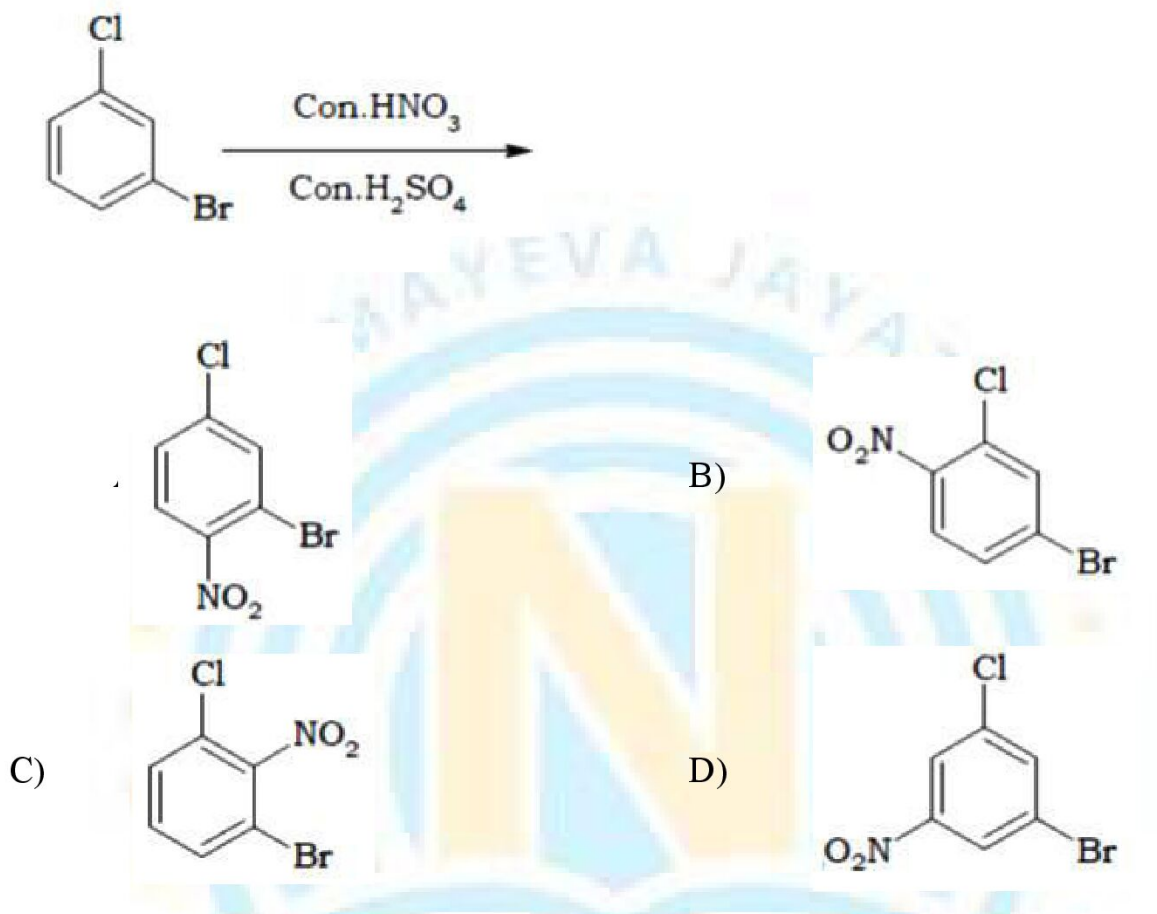
54. The major product of the below reaction is



Key. D

Sol. Electron donating groups increases reactivity of the ring.

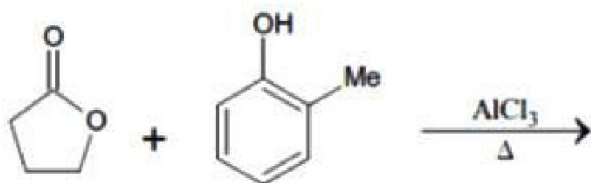
55. Chief product of the following reaction is

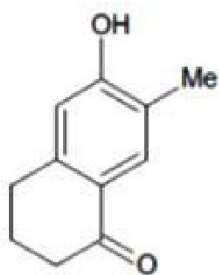


Key. A

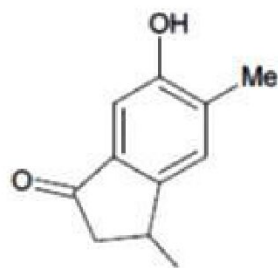
Sol. Both Cl and Br are ortho para directing group

56. The major product of the following reaction is

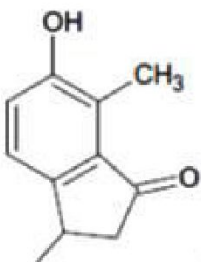




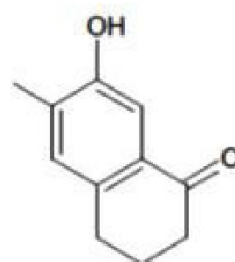
B)



C)

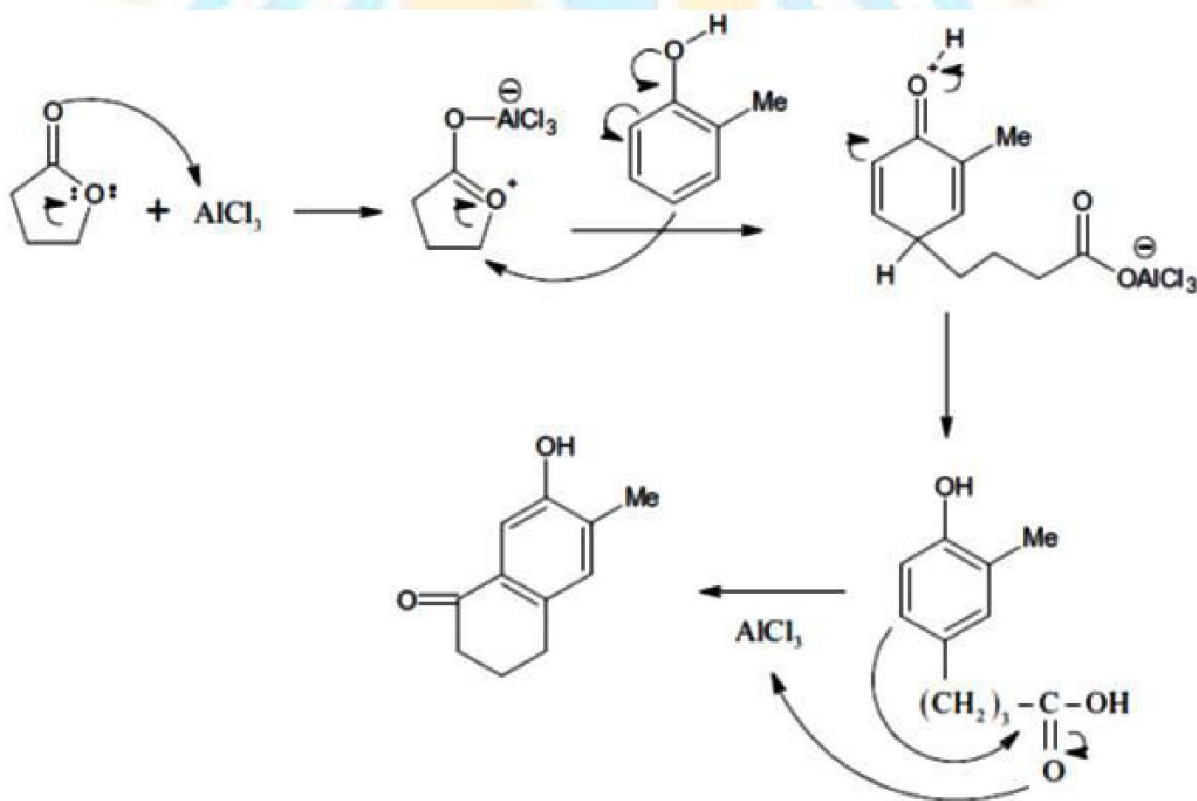


D)



Key. D

Sol.



57. The rate of reactivity of different alkyl halides in Friedel – Craft's reactions

1) $F-I > R-Br > R-Cl > R-F$

2) $R-Cl > R-F > R-Br > R-I$

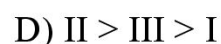
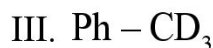
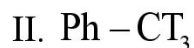
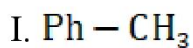
3) $R-F > R-Cl > R-I > R-Br$

4) $R-F > R-Cl > R-Br > R-I$

Key. D

Sol. Reactivity of R – X towards Friedel – Craft's reactions $\propto$ Stability of $[X^{\delta-} \cdots AlCl_3]$ complex

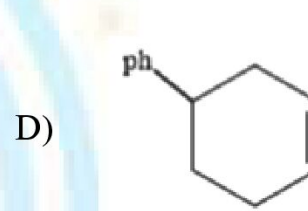
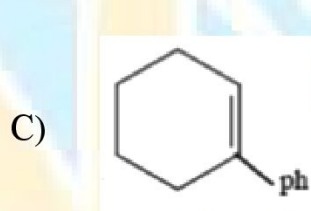
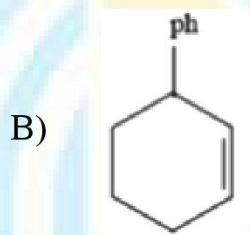
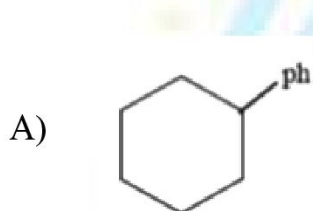
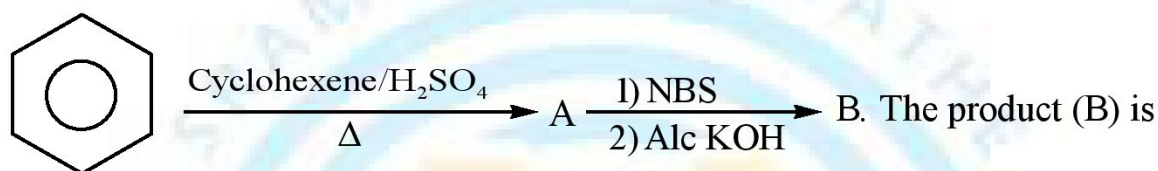
58. Arrange the following in decreasing order of reactivity towards electrophilic aromatic substitution.



Key. 2

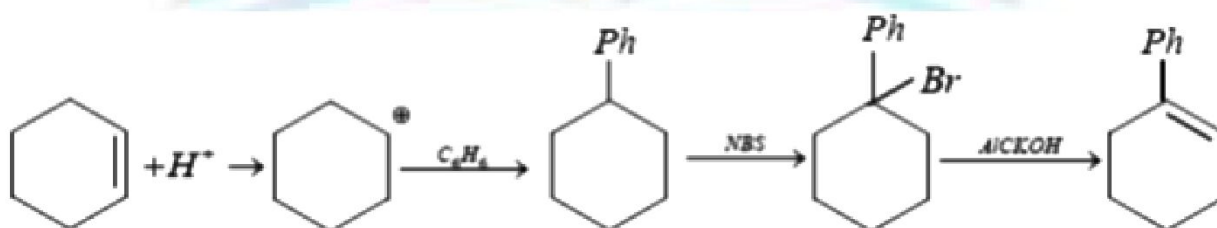
Sol. Order of bond dissociation energy is $C - H < C - D < C - T$

59.



Key. C

Sol.

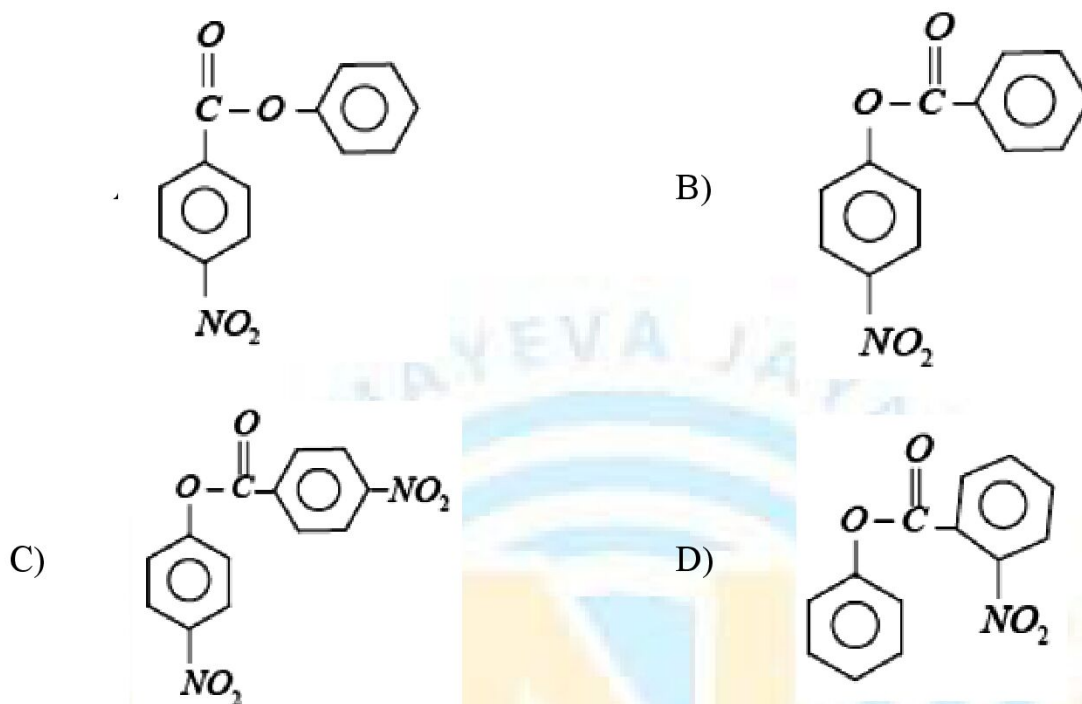
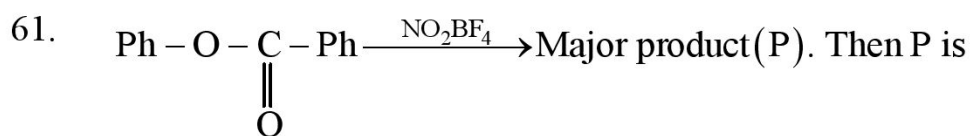


60. Which of the following will not participate in Friedel-Crafts Alkylation reaction?



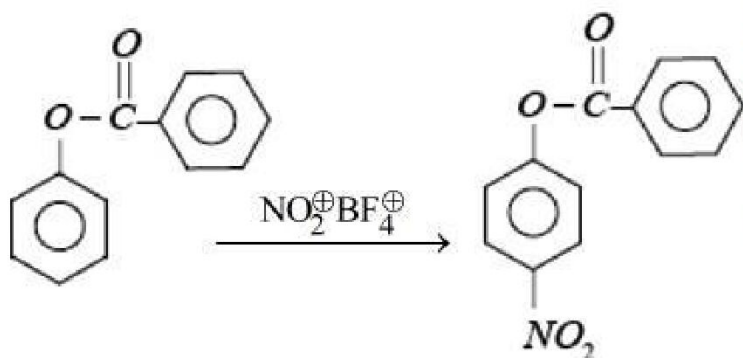
Key. D

Sol. Aniline forms salt with anhydrous $AlCl_3$ and nitrogen benzene, cyanobenzene are less reactive towards EAS.



Key. B

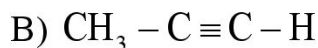
Sol.



Activating groups are strong directing groups towards EAS.



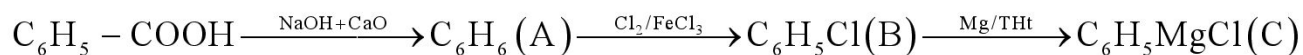
Which of the following reagent is not used to convert the C into A.



C) NMe_3 D) $\text{CH}_3 - \text{CH}_2 - \text{OH}$

Key. C

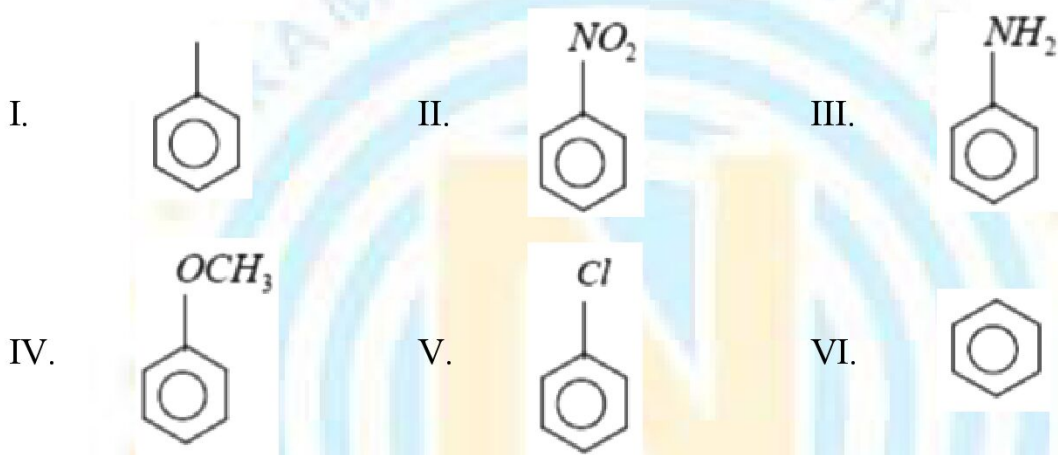
Sol.



X = compounds with acidic hydrogen

 $x = \text{H}_2\text{O}, \text{CH}_3 - \text{C} \equiv \text{CH}, \text{CH}_3 - \text{CH}_2 - \text{OH}$ but not NMe_3

63. Reactivity order of given compounds towards EAS.

A) $\text{I} > \text{III} > \text{V} > \text{VI} > \text{II} > \text{IV}$ B) $\text{III} > \text{IV} > \text{I} > \text{V} > \text{VI} > \text{II}$ C) $\text{III} > \text{IV} > \text{I} > \text{VI} > \text{V} > \text{II}$ D) $\text{IV} > \text{III} > \text{I} > \text{VI} > \text{V} > \text{II}$

Key. C

Sol. Reactivity towards electrophilic aromatic substitution $\propto$ electron density of benzene ring

64. An unknown compound, X, has the molecular formula C_7H_{12} . On catalytic hydrogenation 1 mole of X absorbs 2 moles of H_2 and yields 2-methylhexane. When X is treated with a solution of $\text{Ag}(\text{NH}_3)_2\text{OH}$, a precipitate forms. An analysis of the precipitate shows that it contains silver, and treatment of the precipitate with dilute HNO_3 regenerates X. Which compound represents X?

- A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2 - \text{C} \equiv \text{C} - \text{H}$ B) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_2 - \overset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_3$
- C) $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH} - \overset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_3$ D) $\text{CH}_3 - \overset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_2 - \text{CH}_2 - \text{C} \equiv \text{C} - \text{H}$

Key. D

Sol. Compound X has two degrees of unsaturation since consumes 2 mole H_2 .

Compound X has terminal triple bond since it gives ppt with $\text{Ag}(\text{NH}_3)_2\text{OH}$.

Formation of 2-methylhexane is justified only with option D.

65. Ethane cannot be obtained by the following.

- A) Heating methyl iodide with sodium metal in ether
 B) Hydrogenation of ethene
 C) Sodium acetate on Kolbe's electrolysis
 D) Hydrolysis of Al_4C_3

Key. D

Sol. Al_4C_3 given CH_4

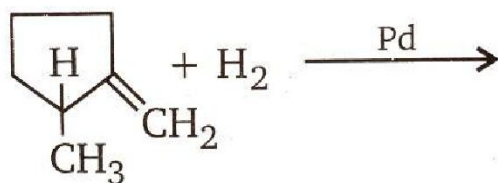
66. Select the **incorrect** statement among the following.

- A) Alkyl halides give alkane with $n\text{-BU}_3\text{SnH}$.
 B) Decarboxylation of formic acid gives methane
 C) Methane is formed from decarboxylation of sodium acetate with NaOH/CaO
 D) Grignard reagent gives alkane with ethyl alcohol.

Key. B

Sol. it gives H_2

67. Product of the below reaction will be.



- A) Racemic mixture
 B) Diastereomers
 C) Meso compound
 D) Constitutional isomers

Key. B

Sol. H_2 , Pd mainly gives syn hydrogenation product

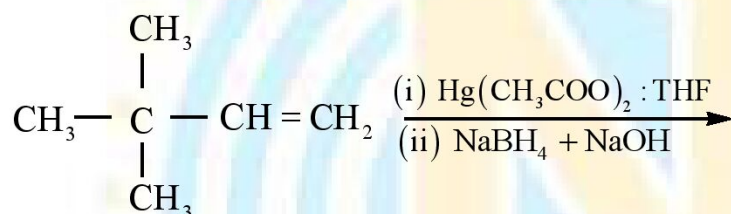
68. Addition of IBr to 2-methylbut-2-ene would form

- A) 2-Bromo-3-iodobutane
 B) 2-Bromo-2-iodobutane
 C) 2-Bromo-3-iodo-2-methylbutane
 D) 2-Iodo-3-bromo-2-methylbutane

Key. C

Sol. Addition of IBr follow Markownikov addition

69. The product of the following reaction is,

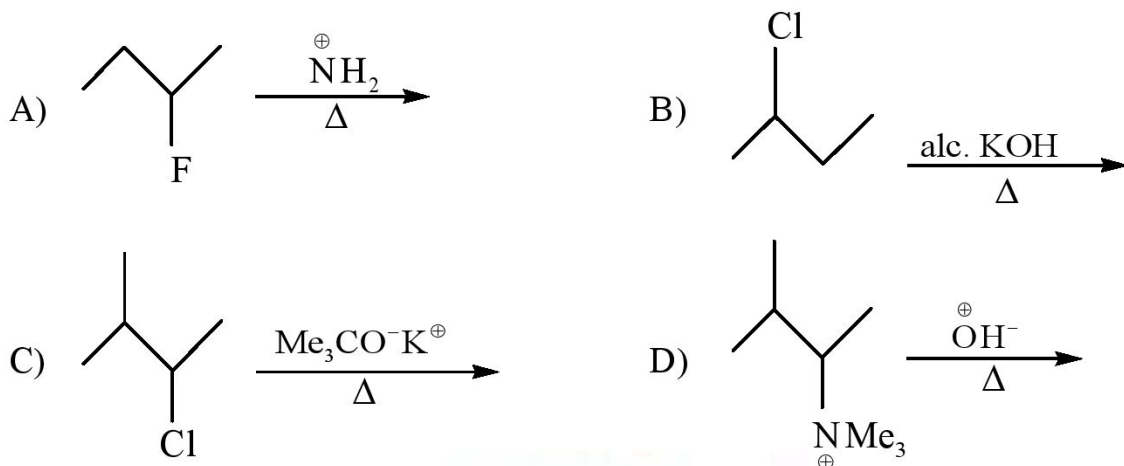


- A) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH} - \text{CH}_3 \\ | \quad | \\ \text{CH}_3 \quad \text{OH} \end{array}$ B) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_2 - \text{CH}_2\text{OH} \\ | \\ \text{CH}_3 \end{array}$
- C) $\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ | \quad | \\ \text{CH}_3 - \text{C} - \text{CH} - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$ D) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{HOCH}_2 - \text{C} - \text{CH}_2 - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$

Key. A

Sol. OMDM is equivalent to markownikov addition of H_2O

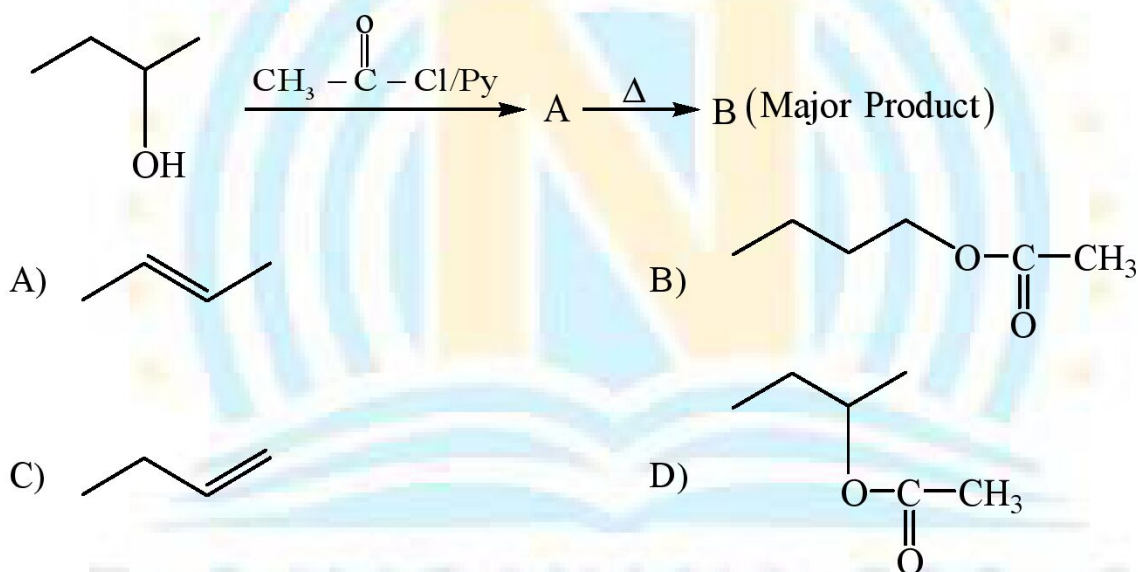
70. Which of the following reaction(s) produce Saytzeff product as a major product.



Key. B

Sol. Alkyl fluorides, bulky bases & Quaternary ammonium salts gives Hoffmann product major product.

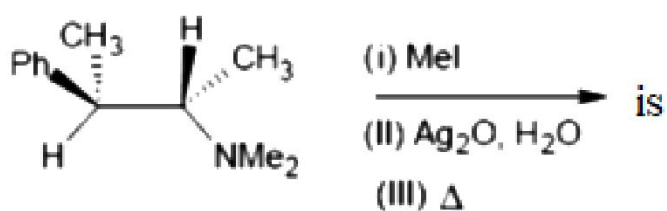
71. The major product B in the above reaction is

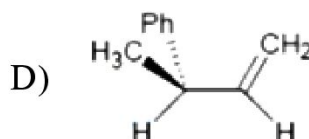
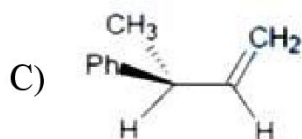
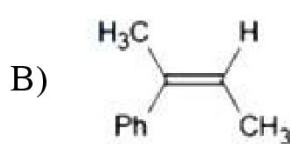
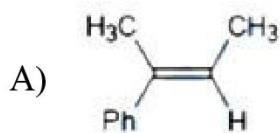


Key. C

Sol. Pyrolysis of esters gives hoffmann's product.

72. The major product obtained in the following reaction.

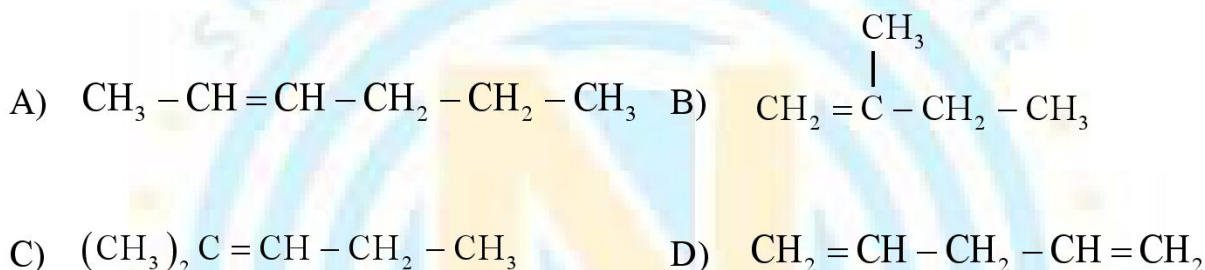




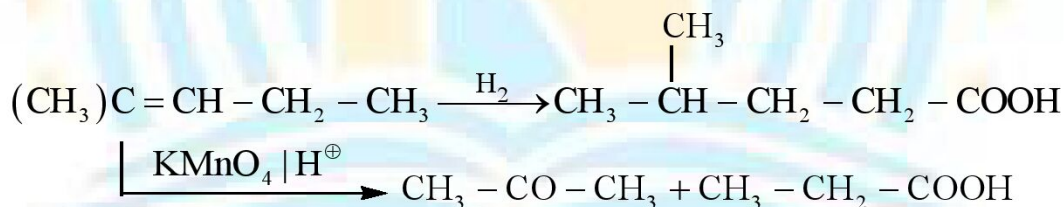
Key. B

Sol. E_2 follows anti elimination.

73. A hydrocarbon 'X' adds on one mole of hydrogen to give another hydrocarbon and decolourised bromine water. 'X' reacts with $KMnO_4$ in presence of acid to give one mole of ketone and one mole of carboxylic acid. The structure of 'X' is

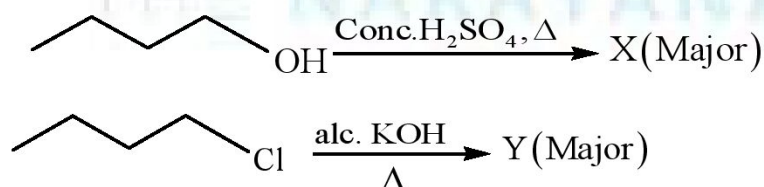


Key. C



Sol.

74. Correct statement among the following.

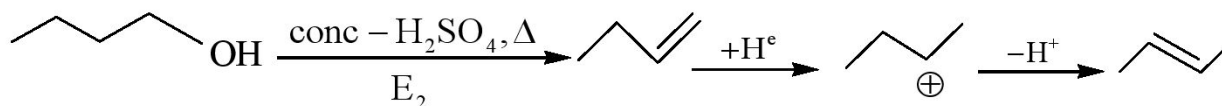


- A) Both X and Y are 1-butene
- B) Both X and Y are trans -2-butene
- C) X is trans 2-butene formed by E_1 mechanism but Y is 1-butene formed by E_2 mechanism.

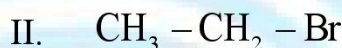
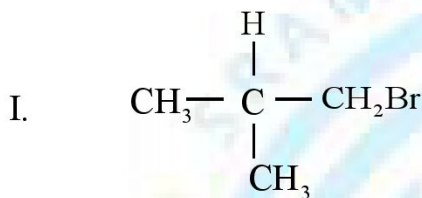
D) X is trans -2-butene formed by E_2 elimination followed by rearrangement, Y is 1-butene formed by E_2 -elimination.

Key. 4

Sol. Less stable alkenes rearrange to more stable alkene by acid catalyst but not by base.



75. Arrange the following alkyl halides in decreasing order of the rate of β -elimination reaction with alcoholic KOH.



A) I > II > III

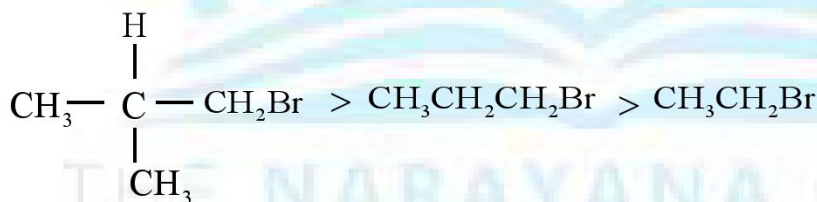
B) III > II > I

C) II > III > I

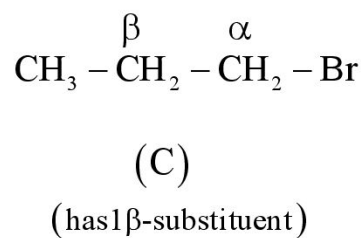
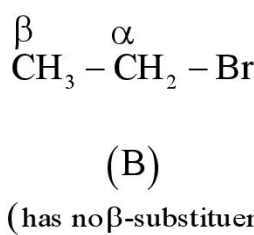
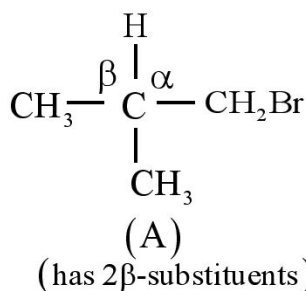
D) I > III > II

Key. D

Sol.



Is the order of rate of β -elimination with alcoholic KOH.

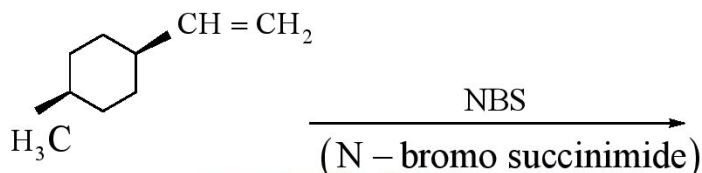


More the number of β -substituents (alkyl groups), more stable alkene will be formed

on β -elimination and more will be the reactivity. Thus, the decreasing order of the rate of β elimination reaction with alcoholic KOH is: $A > C > B$

Integer Type Questions

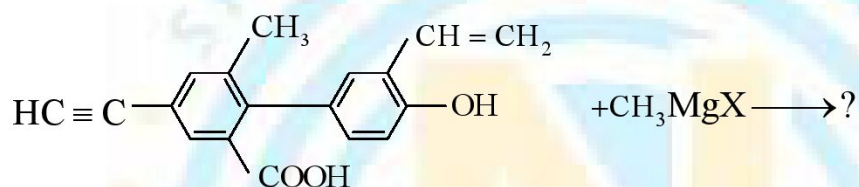
76. Find the total number of possible monobromo products obtained during the reaction.



Key. 4

Sol.

77.

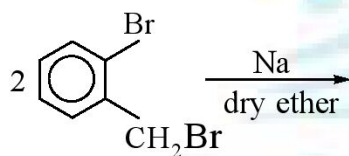


How many mole alkane will be formed?

Key. 3

Sol. There are three acidic hydrogens in reactant that can be abstracted by RMgX

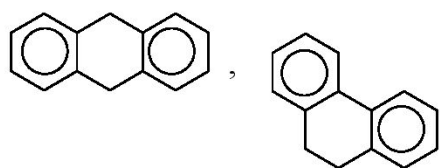
78.



In the above reaction in the number of product (s).

Key. 2

Sol.



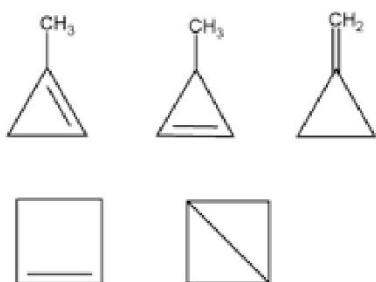
79. If 5-bromo-2-hexene is treated with $\text{Br}_2 / \text{CCl}_4$. How many different tribromides would be formed.

Key. 8

Sol.

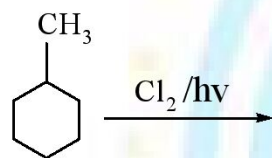
80. The total number of cyclic isomers possible for a hydrocarbon with the molecular formula C_4H_6 is

Key. 5



Sol.

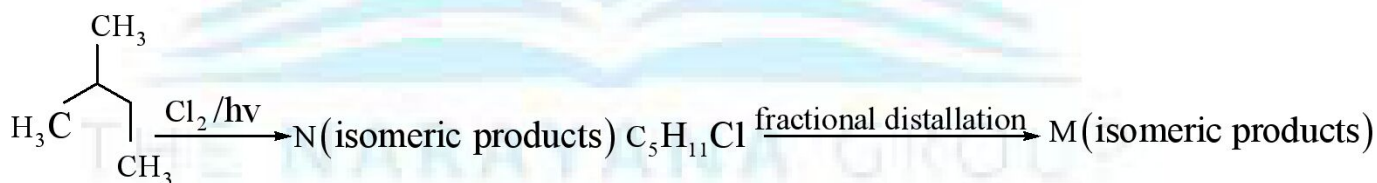
81. Find the total number of isomeric products X in the following reaction and mark $(x - 5)$ in your response sheet.



Key. 7

Sol. Value of X is 12, so ans: 7.

82.

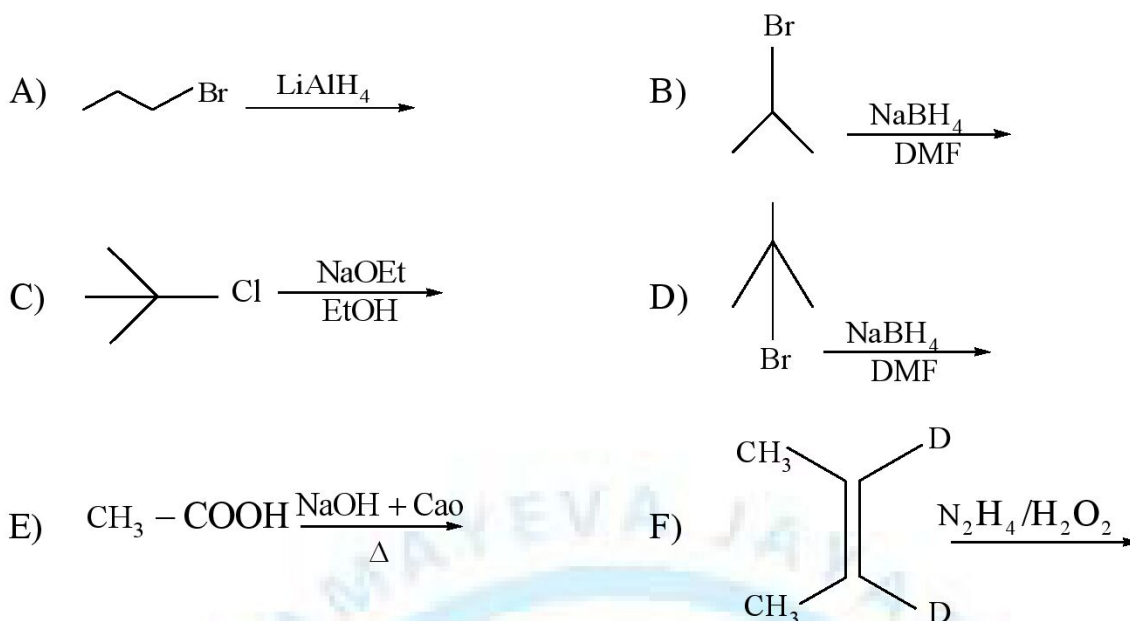


Find the value of M.

Key. 4

Sol. Value of N is 6, that includes two enantiomeric pairs. On fractional distillation four fractions will be obtained. So answer is 4.

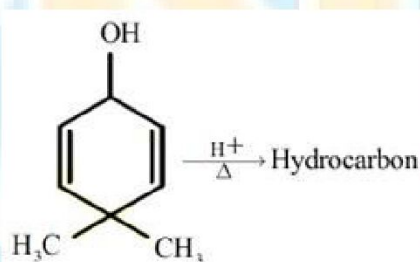
83. Number of reactions in which alkane is formed as major product.



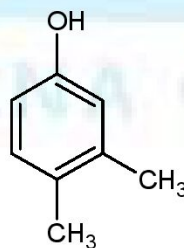
Key. 4

Sol. A, D, E, F produce alkanes

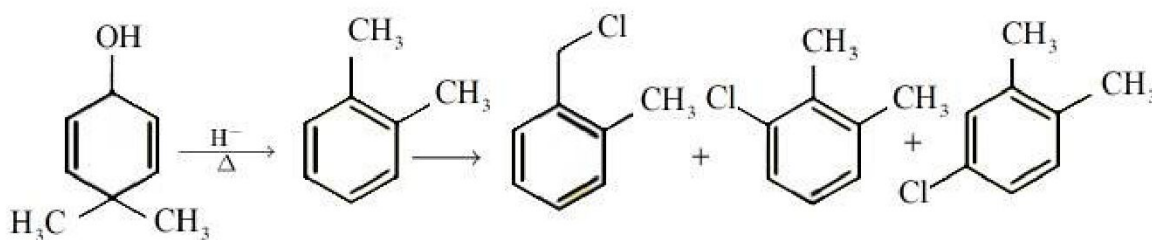
84. Number of mono chloro derivatives of Hydrocarbon is



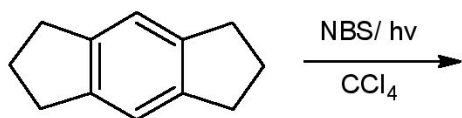
Key. 3



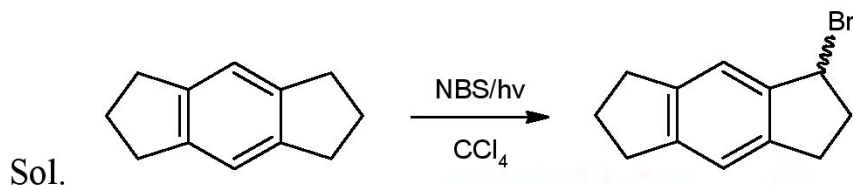
Sol. Hydrocarbon product of given rxn is



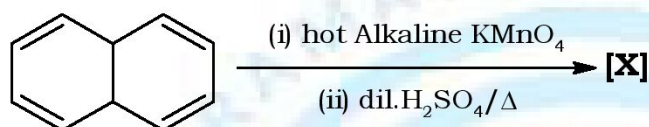
85. Find the number of monobromo derivatives in the given reaction.



Key. 2

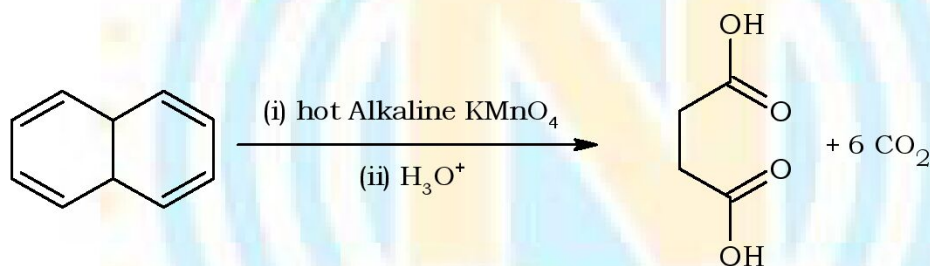


86. The total number of COOH groups in the product [X] is

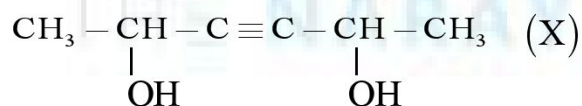


Key. 2

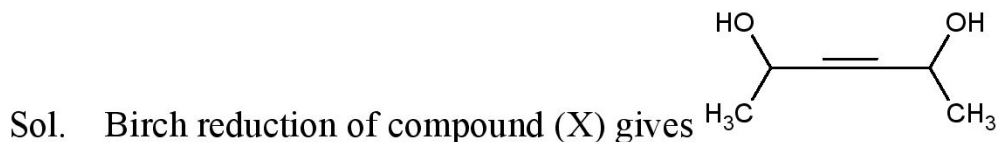
Sol.



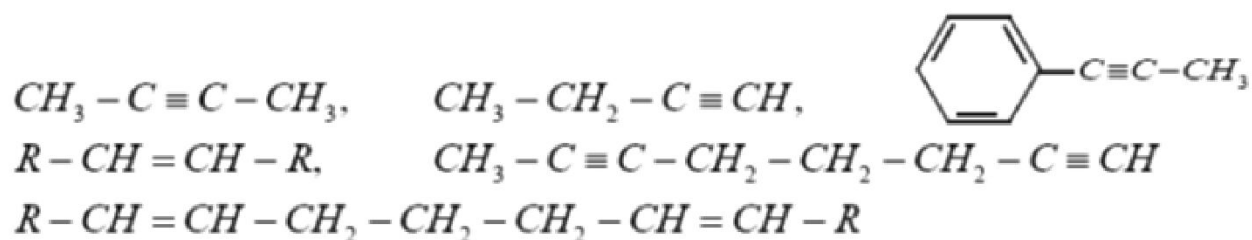
87. Total number of stereoisomers of the product when (X) is partially reduced by either Na/liq NH₃ (Birch reduction) or H₂, Pd -BaSO₄ (consider only double bond reduction) are



Key. 6



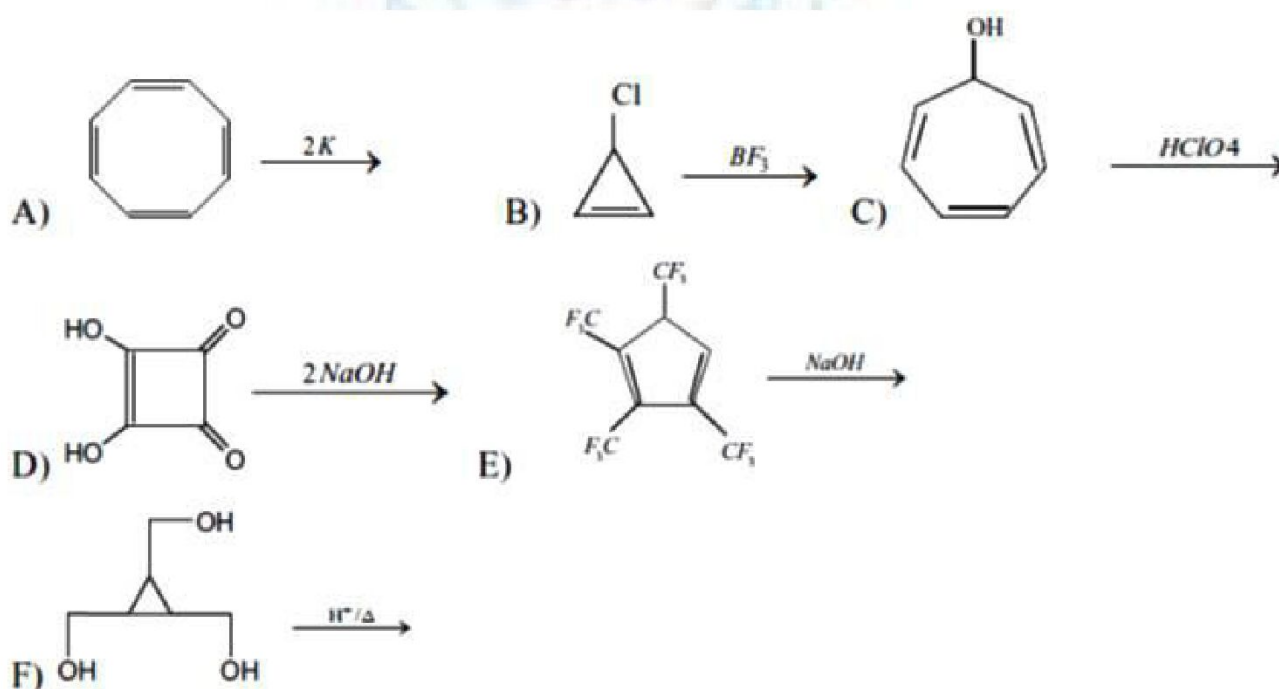
88. How many of the following compounds are reduced by Li in liq. NH₃ to alkene



Key. 2

Sol. Only compound 1 and 3 are reduced to alkenes

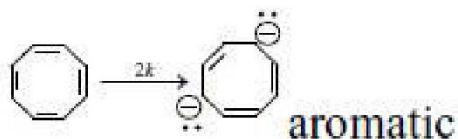
89. The number of reactions among the following that produce aromatic compounds is



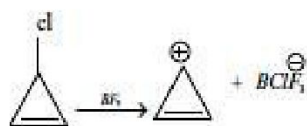
Key. 6

Sol.

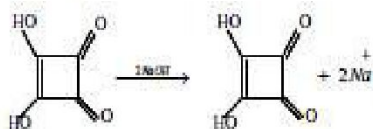
A)



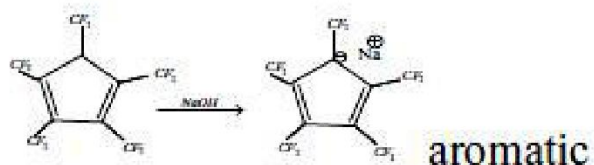
B)



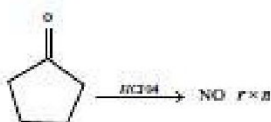
C)



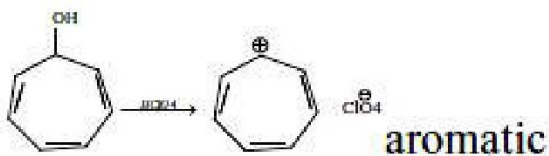
D)



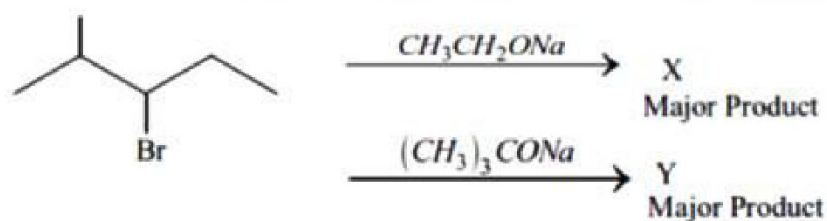
E)



F)



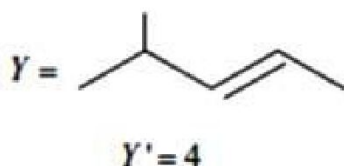
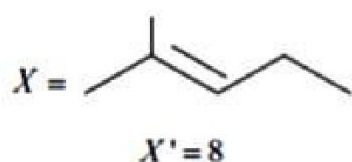
90.



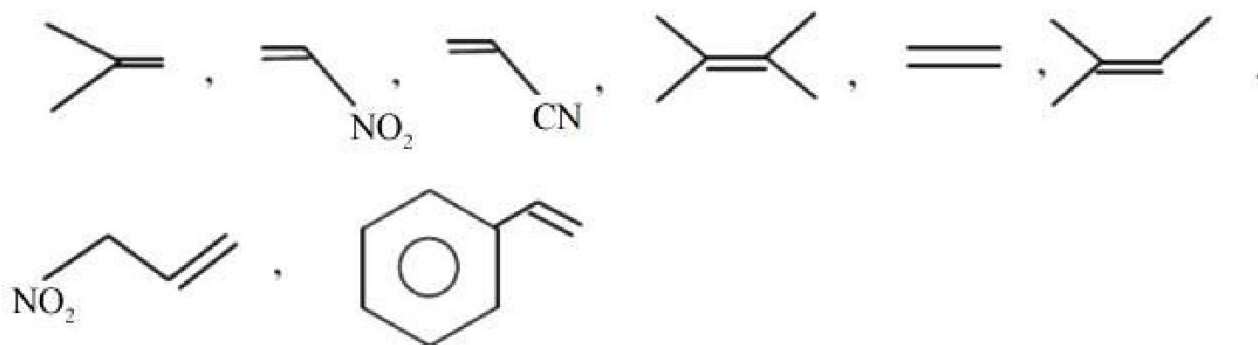
If x is the number of hydrogens involved in hyper conjugation in 'X' and y is the number of hydrogens involved in hyper conjugation in 'Y', then the value of $(x-y)$ is

Key. 4

Sol.

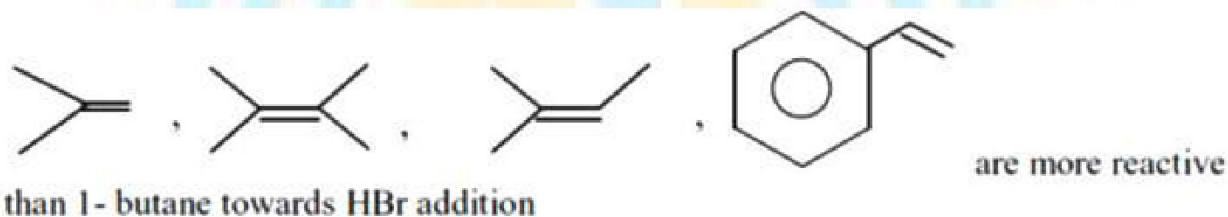


91. How many of the following alkenes are more reactive than but-1-ene towards addition of HBr?



Key. 4

Sol.



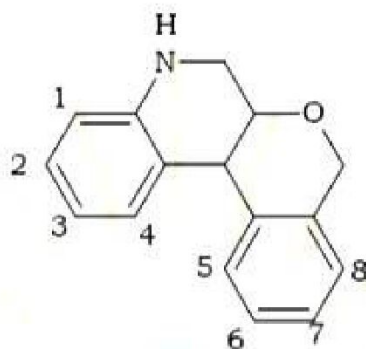
92. How many of the following reagents are involve in syn addition with alkene?

- (i) $\text{Br}_2 / \text{CCl}_4$ (ii) H_2 / Ni (iii) $\text{Li} / \text{Liq. NH}_3$ (iv) Tilden's reagent
 (v) $\text{PhCO}_3\text{H} / \text{H}_2\text{O}$ (vi) Baeyer's reagent
 (vii) $\text{N}_2\text{H}_4 / \text{H}_2\text{O}_2$ (viii) $\text{Si}_2\text{BH} / \text{H}_2\text{O}_2 / \text{OH}^-$

Key. 4

Sol. H_2 / Ni , Bayer's reagent, $\text{N}_2\text{H}_4 / \text{H}_2\text{O}_2$, $\text{Si}_2\text{BH} / \text{H}_2\text{O}_2 / \text{OH}^-$

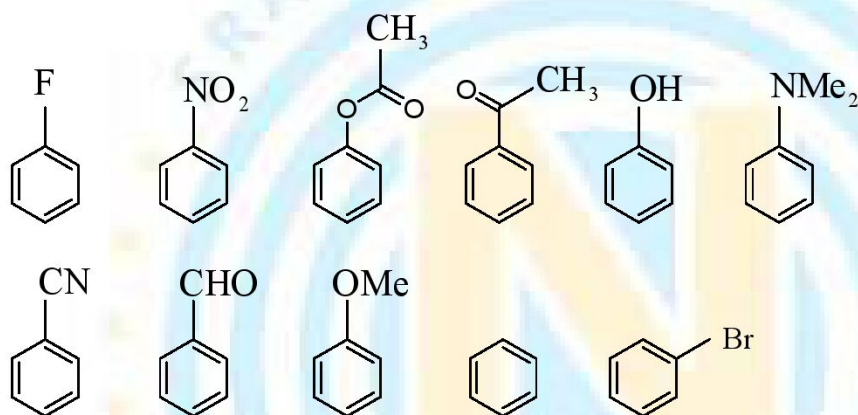
93. From which position does NO_2^+ replace a hydrogen from the following compound predominately?



Key. 3

Sol.

94. How many of the following compounds are more reactive than chlorobenzene towards nitration?

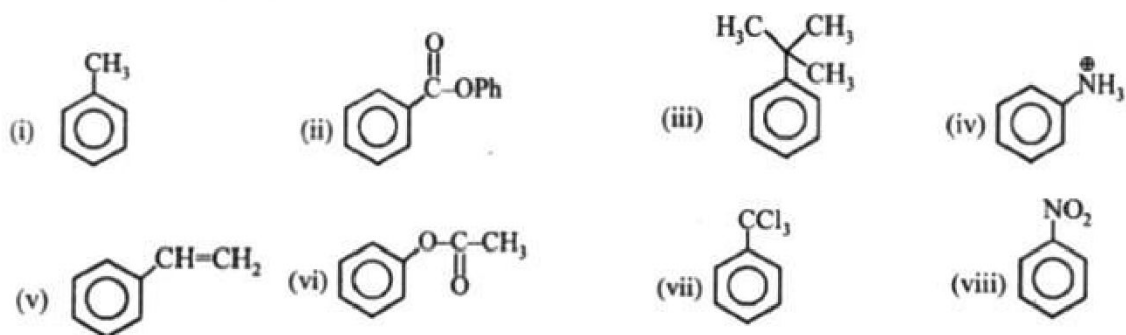


Key. 6

Sol.



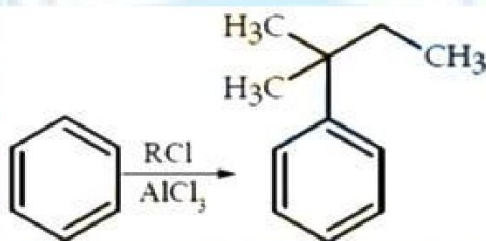
95. Total number of compounds which on mono nitration gives ortho or para substituted major product.



Key. 6

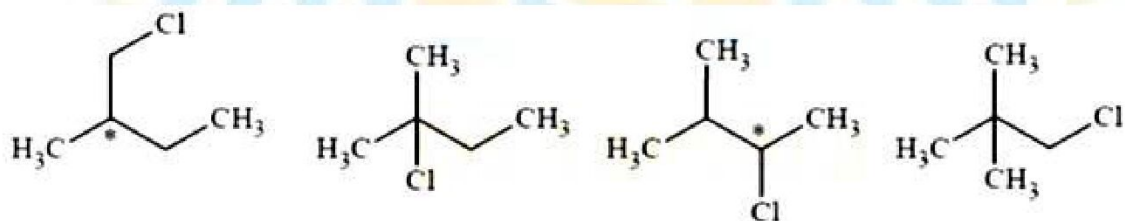
Sol.

96. Find the total number of possible alkyl chloride in the following reaction.

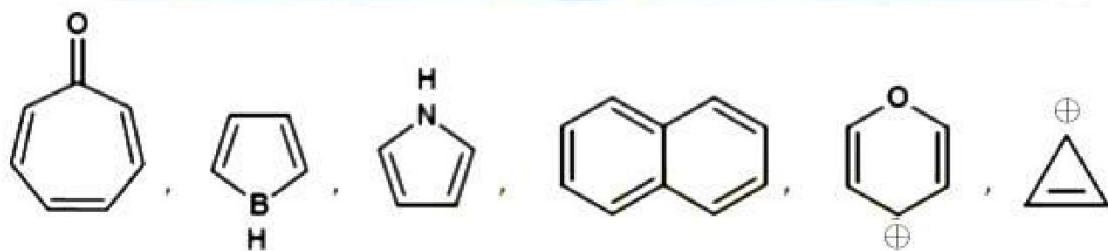


Key. 7

Sol.

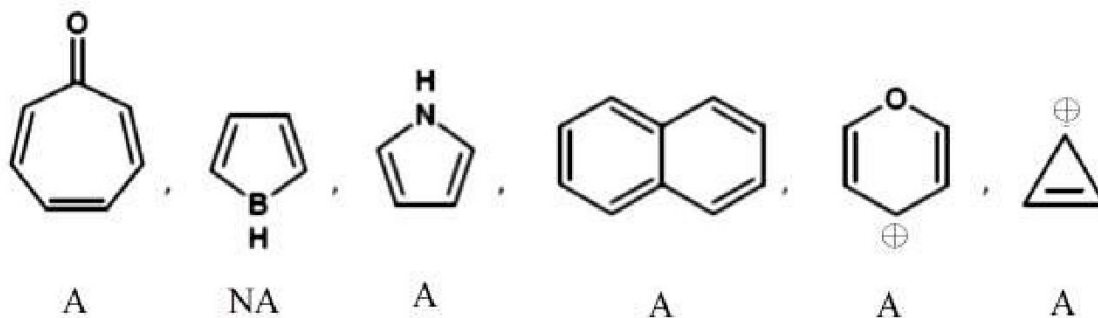


97. How many structure(s) out of the following are aromatic.

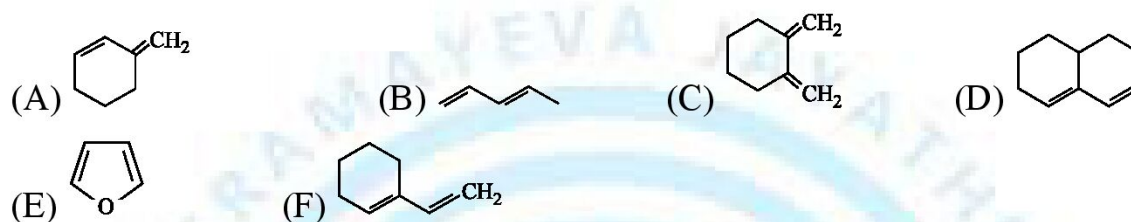


Key. 5

Sol.



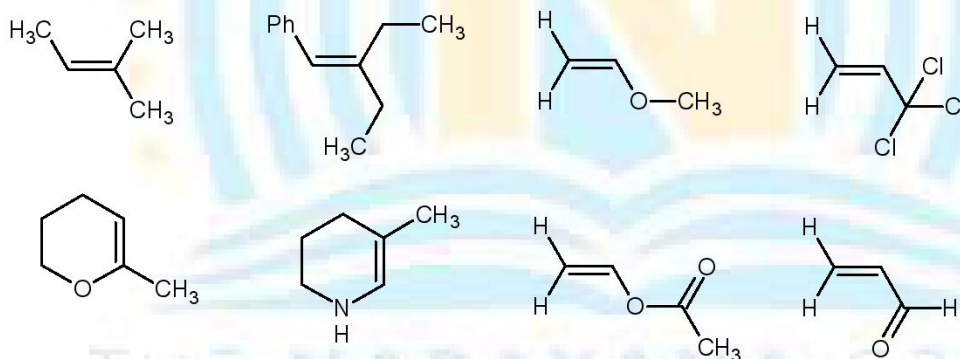
98. How many of the following conjugated diene would not react with a dienophile in a Diels–Alder reaction?



Key. 2

Sol. A,D

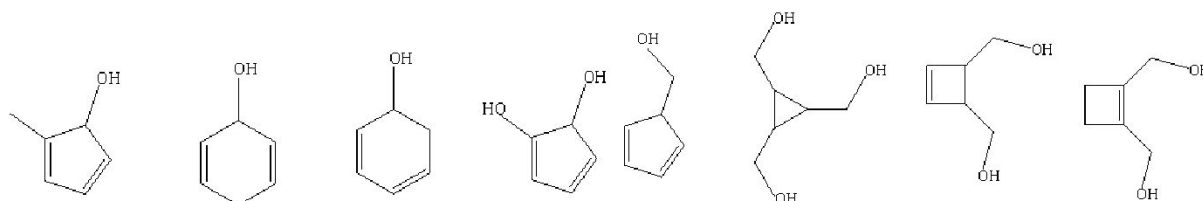
99. How many of them will give Markovnikov product as a major during addition of HBr?



Key. 4

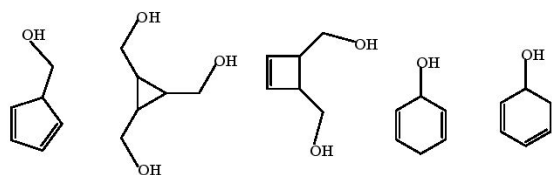
Sol. Compound 1, 3, 5, 6 give Markovnikov product

100. How many of the following compounds on dehydration with Conc. H_2SO_4 gives benzene as major product?

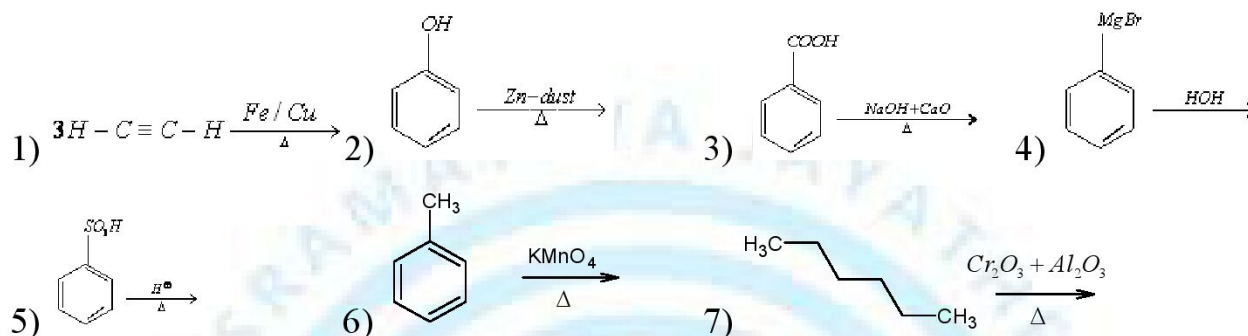


Key. 5

Sol.



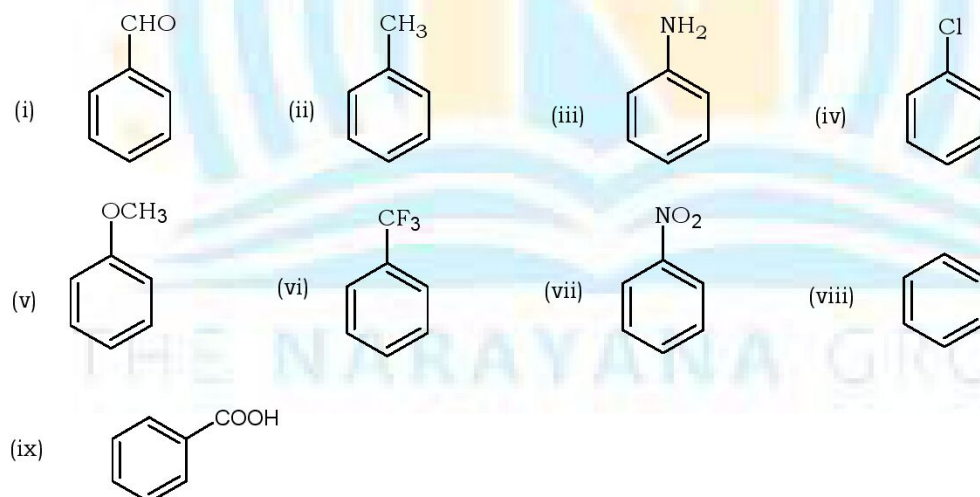
101. How many of the following reactions gives benzene as major product?



Key. 6

Sol. Reaction 1, 2, 3, 4, 5, 7 gives benzene as major organic product

102. The number of compounds which do not undergo Friedel – Crafts acylation is



Key. 5

Sol. Compound (i), (iii), (vi), (vii), (ix) do not undergo Friedel – Crafts acylation



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